



Developer's Guide

Informed by MCG

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1 Getting started

Use the Informed by MCG for Utilization Management web service to integrate the evidence-based care guidelines into your medical management or other software solution. With this web service, you can perform a variety of functions using licensed care guideline content while employing your own user interface and user experience.

The guidelines exposed by our web services are provided as is and shall be presented to the user in their entirety. It is not permissible to decompose or otherwise map clinical concepts against our guidelines or indications within our guidelines without the express permission of MCG. Without explicit permission for such mapping MCG will not certify an integration for use.

1.1 Prerequisites

To use the Informed by MCG for Utilization Management web service, you must obtain licenses for each web service and MCG guideline product you intend to use. To assist you in obtaining the required licenses, contact your MCG account executive.

1.2 API versions

As MCG continues to enhance the Informed by MCG web services, newer versions may be introduced. New features may not be included in previous versions. Integrating solutions are encouraged to use the latest version available at the time of integration and update as new versions are made available.

1.3 Custom guideline support

The Informed by MCG for Utilization Management web service supports customized guidelines created using the Guideline Modification Module. Support is limited to custom guideline sections with titles that match MCG guideline section titles.

1.3.1 Rich text formatting

With the **richtext** parameter in API Version 2 of the endpoints, all formatting (such as bold text, italicized text, font type, and color) applied to a section's top-level paragraph or top-level indication in Guideline Modification Module can be displayed as HTML ** tags in the response. Previously, formatted text appeared as plain text only. See [Optional parameters](#) for details.

1.4 Guideline editions

One or more editions of the MCG care guidelines are published each year. Guideline versions are designated as "ee.v" where ee is the overall guideline edition, such as 28, and .v is the version within that edition, such as .1 or

.2. For example, if two versions of the 28th Edition are released, their version numbers would be 28.0 (initial) and 28.1 (second version).

You can specify which version of the care guidelines that you want to work with in each request submitted using the Informed by MCG for Utilization Management web service. If no version is specified, Informed by MCG for Utilization Management uses the latest available version.

1.5 Guideline product abbreviations

You may see all of these guideline products or a subset of them at your site. For more information about the guidelines that are available for you, contact your application administrator.

Abbreviation	Guideline product name
AC	Ambulatory Care
BHG	Behavioral Health Care
GRG	General Recovery Care
HC	Home Care
ISC	Inpatient & Surgical Care
MCM	Multiple Condition Management
ORG	Optimal Recovery Guidelines
MCR	Medicare Compliance
PIP	Patient Information
RFC	Recovery Facility Care

1.6 Guideline types

The following list describes the MCG guideline types that commonly appear.

Product	Guideline type	Type description
AC	ACG	Ambulatory Care Guideline
AC	RMG	Referral Management Guideline
BHG	HC	Home Care Guideline
BHG	LOC	Level of Care Guideline
BHG	ORG	Optimal Recovery Guideline
BHG	PDI	Patient Discharge Information
BHG	RFC	Recovery Facility Care Guideline
BHG	TTP	Therapeutic, Testing and Community Service Guideline

Product	Guideline type	Type description
GRG	ADV	GRG Advisory Category
GRG	CM	Case Management Guideline
GRG	GRG	General Recovery Guideline
GRG	LOC	Level of Care Guideline
GRG	LTACH	Long Term Acute Care Hospital Guideline
HC	GRG	General Recovery Guideline
HC	ORG	Optimal Recovery Guideline
ISC	CCC	Common Complication and Condition Guideline
ISC	LOC	Level of Care Guideline
ISC	ORG	Optimal Recovery Guideline
ISC	ORG-AP	Optimal Recovery Guideline, Adult and Pediatric
ISC	ORG-P	Optimal Recovery Guideline, Pediatric
ISC	RRG	Rapid Review Guideline
ISC	RRG-AP	Rapid Review Guideline, Adult and Pediatric
ISC	RRG-P	Rapid Review Guideline, Pediatric
MCM	ORG	Optimal Recovery Guideline
MCR	NCD LCD	National Coverage Determination Local Coverage Determination
PED	PTED	Patient Education
PED	PTED-SP	Patient Education, Spanish
PIP	ICP	Inpatient Care Plan
PIP	PDI	Patient Discharge Information
PIP	PREOP	Preoperative Information
RFC	GRG	General Recovery Guideline
RFC	IRF	Inpatient Rehabilitation Facility Guideline
RFC	ORG	Optimal Recovery Guideline

1.7 Annotation types

The following list describes the annotation types that commonly appear in guidelines.

Annotation type	Type description
ALT	Alternatives in BHG, ISC, and RFC
CRI	Criteria in BHG, ISC, and RFC

Annotation type	Type description
EVI	Evidence in CCG
HOS	Hospitalization in ISC
LOC	Level of Care in BHG and ISC
LOS	Length of Stay in BHG, ISC, and RFC
OPS	Operative Status in ISC
REMREL	Remission and relapse in BHG and ISC
TREAT	Treatment in BHG and ISC

1.8 Available resources

- The MCG Swagger web service test site. This is the primary source of information about MCG request and response schemas. Using this site, you can model requests and try out the Informed by MCG for Utilization Management endpoints.
- Informed by MCG for Utilization Management Implementation Guide. This guide is intended for all audiences. It describes the web service's primary business use cases, minimum installation requirements, and marketing and branding requirements for MCG guideline content that you display in your own solution. It also includes a certification checklist that describes the minimum functionality to be implemented by the vendor partner.

MCG Technical Support is available Monday - Friday, 5:00 AM to 5:00 PM Pacific Time. Calls outside of the 5:00 AM to 5:00 PM window are routed to an answering service. For critical issues impacting multiple users (for example, production system down), the answering service will contact the MCG Technical On-Call Support Agent who will return your call. They can be reached by telephone at 800 598 2292, or by email through the MCG.com website: <https://www.mcg.com/contact-us>.

1.9 About the best practices information



Throughout this document you will find information about best practices for implementing and using the Informed by MCG for Utilization Management web service. These best practices are called out with the community icon appearing to the left of this paragraph.

2 Optional parameters

This following optional parameters may be used.

2.1 LCD region

Local Coverage Determination (LCD) guidelines are available in the Medicare Compliance (MCR) content product. The optional parameter **lcdRegions** is included with the Search endpoints. You can enter multiple values to locate LCD guidelines for one or more geographic regions.

2.2 MAO parameter

The GET and POST Indication and Authorization endpoints offer integrated solutions the option to include the Supplemental Medicare Criteria section for Inpatient & Surgical Care (ISC) and General Recovery Care (GRG) guidelines in the response for display to Medicare Advantage Organizations (MAOs). This can be done by setting the optional MAO parameter either to True or False from the endpoints. This section appears after the Clinical Indications for Admissions section and is used to evaluate all criteria required for the POST Met or Not Met calculation.

Care Planning - Inpatient Admission and Alternatives

[Return to top of Stroke: Hemorrhagic - ISC](#)

Clinical Indications for Admission to Inpatient Care

[Return to top of Stroke: Hemorrhagic - ISC](#)

- Admission is indicated for 1 or more of the following(1)(2)(3)(4)(5)(6)(7):
 - Acute hemorrhagic (eg, intracerebral) stroke

Supplemental Medicare Criteria

[Return to top of Stroke: Hemorrhagic - ISC](#)

Note: These criteria are to be considered for Medicare patients if none of the standard Clinical Indications for Admission to Inpatient Care apply.

- Patient with Medicare coverage requires inpatient admission, as indicated by 1 or more of the following(9):
 - Admitting clinician expects patient to require hospital care for less than 2 midnights but, based on complex medical factors documented in medical record
 - Patient has need for intubation and mechanical ventilation that is new (ie, did not present to hospital already on mechanical ventilation).
 - Treatment plan for hospital admission includes procedure designated by CMS as inpatient only (ie, on Inpatient Only List).
 - Patient has already received medically necessary hospital-level care that meets 2-midnight benchmark (excluding activities such as triage/intake, del

The Supplemental Medicare Criteria section (MAO content) is included for applicable guidelines when set to True.

mao	true	Whether to return MAO content
	true false (default)	

2.2.1 Example: mao = True

The following example shows how the response for the GET Indications endpoint displays the Supplemental Medicare Criteria section after the Clinical Indication for Admission section within the M-85 guideline content structure when set to True.


```
{
  "Sections": [
    {
      "Heading": "Clinical Indications for Admission to Inpatient Care",
      "Content": [
        {
          "Text": "Admission is indicated for 1 or more of the following[[Cit:2714572]]
[[Cit:189017]][[Cit:2713170]][[Cit:2713137]][[Cit:252425]][[Cit:245931]][[Cit:2543836]]:[[Ano:1]]",
          "ListLogic": "1 or more of the following",
          "Id": "8DE994F59626DF23",
          "PathId": "8DE994F59626DF23",
          "AnnotationLinks": [
            {
              "Id": "1",
              "AnnotationType": "CRI",
              "Link":
"https://EndpointURL.com/api/v2/GA/27.2/ISC/0010224600b/annotation/1"
            }
          ],
          "CitationLinks": [
            {
              "Id": "2714572",
              "DisplayValue": "1",
              "Link": "https://EndpointURL.com/api/v2/GA/27.2/citation/2714572"
            }
          ],
          ...
        }
      ],
      {
        "Heading": "Supplemental Medicare Criteria",
        "Paragraph": "These criteria are to be considered for Medicare patients if none of the
standard Clinical Indications for Admission to Inpatient Care apply.",
        "Content": [
          {
            "Text": "Patient with Medicare coverage requires inpatient admission, as indicated
by 1 or more of the following[[Cit:2814737]]:",
            "ListLogic": "1 or more of the following",
            "Id": "CF7468E49DC1BB2F",
            "PathId": "CF7468E49DC1BB2F",
            "CitationLinks": [
              {
                "Id": "2814737",
                "DisplayValue": "1",
                "Link": "https://EndpointURL.com/api/v2/GA/27.2/citation/2814737"
              }
            ],
            "ContentItems": [
              {
                "Text": "Admitting clinician expects patient to require hospital care for
less than 2 midnights but, based on complex medical factors documented in medical record, judges that
inpatient care is necessary (case-by-case exception).",
                "Id": "06A8E5D0597EFFFFC",
                "PathId": "CF7468E49DC1BB2F-06A8E5D0597EFFFFC"
              },
              {
                ...
              }
            ]
          }
        ],
        ...
      }
    ]
  }
}
```

When `mao = True`, the POST Indication endpoint lets you return the IDs and criteria to get back a Met or Not Met status. Ensure that the MAO parameter is set to True when calling the POST endpoint and indicating IDs within the Medicare Advantage supplemental subsection. This is necessary for the supplemental Medicare data to be considered in the Met or Not Met evaluation. If the MAO parameter is not set to True, the supplemental Medicare criteria IDs will be ignored.

2.2.2 Example: `mao = False`

By default, the MAO content is not displayed within the guideline content.

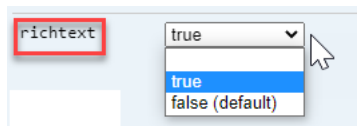
If `mao = True` but no supplemental data exists for the guideline, the response will be the same as when requesting `mao = False`.

2.3 Richtext parameter

In API Version 2, the **richtext** parameter lets you display all occurrences of formatted text (including bold text, italicized text, font type, and color) within the guideline information as HTML `<span>` tags in the response. Previously, formatted text appeared as plain text only.

Also, for rich text, emphasis, and bold or red formatting to be consistent, the response will display all reference types (including citation, footnote, quality measure, and annotation) as HTML links. Previously, references appeared as plain text inside `[[double brackets]]`.

To display rich text in the response, select True as shown.



2.3.1 Example: `richtext = True`

The following example shows how the response for the Indication endpoint returns **bold** text in a logic rule, and additionally, how the HTML links for references (citation, annotation) display when `richtext = True`.

```
{
  "Heading": "Clinical Indications for Procedure",
  "Content": [
    {
      "Text": "Procedure is indicated for >ALL</span> of the
following<a class='citation' data-id='2538537'>1</a><a class='citation' data-id='2540611'>2</a><a
class='citation' data-id='188363'>3</a><a class='annotation' data-id='2'></a>",
      "ListLogic": "All of the following",
      "Id": "3C0C1498C4210858",
      "PathId": "3C0C1498C4210858",
      "AnnotationLinks": [
        {
```

2.3.2 Example: richtext = False

The following example shows how the response returns rich text formatting as plain text, and additionally, all reference types (citation, annotation) inside double brackets when richtext = False.

```
{
  "Heading": "Clinical Indications for Procedure",
  "Content": [
    {
      "Text": "Procedure is indicated for ALL of the following[[Cit:2538537]][[Cit:2540611]]
[[Cit:188363]]: [[Ano:2]]",
      "ListLogic": "All of the following",
      "Id": "3C0C1498C4210858",
      "PathId": "3C0C1498C4210858",
      "AnnotationLinks": [
        {
          ...
        }
      ]
    }
  ]
}
```

The response for each reference type is shown in the following table.

Reference type	rich text = true	rich text = false
Citation	{citationNumber}	[[Cit:citationId]]
Footnote Ref / Footnote	{footnoteLetter}	[[Foot:footnoteId]]
Quality Measure	QM	[[qualityMeasure-QM]]
Annotation		[[Annotation:annotationId]]

3 Authentication and headers

Your solution is responsible for performing authentication, including validating user login identity and managing all other front-line security. Informed by MCG for Utilization Management web service uses the following items to validate each request:

- A client ID provided by MCG that uniquely identifies a client.
- A client key provided by MCG that is unique to a client ID.
- A calculated hash value that is unique to each GET and POST request.

Partners that are integrating with Informed by MCG for Utilization Management for commercialization are provisioned with a single client ID and key to use in their test environment for development purposes. Clients will be provisioned with both test and production credentials.

Use the SHA-256 algorithm for all hash calculations, as shown in the following section. The hash value must be calculated uniquely for each GET or POST message your application sends to Informed by MCG for Utilization Management.

Your site must be enabled for https to connect with the MCG site.

3.1 Individual message integrity

The message header for each GET or POST request sent to Informed by MCG for Utilization Management must contain your client ID and a hash value calculated from your client ID, the client key, and the web service endpoint.

After receiving a request, the Informed by MCG for Utilization Management web service calculates its own hash value for the request, and compares the values. The URI-encoding of the https request happens first, and the results are used for signature generation/verification. If the hash values do not agree, the message is considered to have been tampered with, and is rejected. If the hash values do agree, the Informed by MCG for Utilization Management web service assumes that the entire message is valid.

The following example shows a hash value calculation and the resulting request to the Informed by MCG for Utilization Management web service.

```
<?php
// Client Identity
$clientID = 'DEMO';
$apiKey = 'ABCDEFGHJKLMNOPQRSTUVWXYZ';
// Service Details
$url = 'https://service.url';
$endpoint = '/api/bundle';
$uri = $url . $endpoint;
$method = 'GET';
// Calculate the message digest hash
$dataToEncode = $endpoint . $clientID . $method . $apiKey;
$hashedData = hash('sha256', $dataToEncode, true);
$encodedHashedData = base64_encode($hashedData);

// Call the Service
$context = stream_context_create(
    array(
```

```
'http' => array(
    'method' => $method,
    'header' =>
        'ClientId: ' . $clientId . "\r\n" .
        'Signature: ' . $encodedHashedData . "\r\n" .
        'Content-Type: ' . 'application/json',
    )
);

$response = file_get_contents($uri, false, $context);
header('Content-Type: application/json');

// Display the response
echo $response;
?>
```

3.2 Partner information in message headers

The following partner-specific information can be included in the message header for each GET or POST request sent to Informed by MCG. This action is optional, but will be required for 2024 certifications.

- Partner identifier: Mcg-Partner-ID
- Solution identifier: Mcg-Solution-ID
- Solution version: Mcg-Solution-Version

These headers must be sent through the partner integration with Informed by MCG. For details on the expected values, contact your MCG account executive.

3.2.1 Partner Identifier

The Mcg-Partner-ID header value is made up of two parts; a prefix and a suffix, concatenated together using the pipe “|” character. The prefix is a UUID (universally unique identifier) that is unique to each partner, and which is also provided to the partner by MCG. The suffix is a string value that is the name of the partner. This name is “free-form” and can be set at the partner’s discretion, assuming that formatting guidelines are followed.

The format of the Mcg-Partner-ID is as follows: <MCG provided UUID>|<Partner supplied name>.

For example, this value may be “00000000-0000-0000-0000-000000000000|Awesome_MCG_Partner_LLC”

The prefix is formatted as: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX. Once provided to the partner, this value should never change.

The suffix portion of the Mcg-Partner-ID can be set to any value - assuming it follows formatting guidelines - at the partner’s discretion. This value is typically the business name.

The formatting rules are:

- Alpha-numeric values, including the underscore character (_) and the hyphen (-) or dash.
- Up to 25 characters (anything more is truncated).
- No white space

- No punctuation, such as but not limited to periods, commas, semi-colons, and so on.
- No special characters, for example, single and double quotes, ampersand (&), carat.



You can use this regular expression (“regex”) to validate that the value being assigned to the field Mcg-Partner-ID meets formatting requirements:

```
^[0-9A-Fa-f-]{36}\\[A-Za-z0-9_\\-]{1,25}$
```

3.2.2 Partner solution identifier

The Mcg-Solution-ID header can be set to any value at the partner’s discretion. This value is typically the commercial name of the application integrated with Informed by MCG. For example, this value may be “CommercialApplicationName”.

3.2.3 Partner solution version

The Mcg-Solution-Version header can be set to any value at the partner’s discretion. This value is typically the version of the application integrated with Informed by MCG. For example, this value may be “1.0.3”. This value should change as the shared client’s version of the application changes.

4 Informed by MCG for Utilization Management endpoints

Many of the Informed by MCG for Utilization Management endpoints require a version/edition number, such as for the 28th Edition. You can use three formats:

Format	Example
<i>nn</i>	28
<i>nn.nn</i>	27.1 or 27.10
latest	latest

4.1 Commonly used endpoints

The following endpoints are common to all Informed by MCG solutions.

- Guideline
- Ping
- Product
- Search
- Version

The following endpoints may be used to retrieve related guideline information and evidence:

- Annotation
- Citation
- Definition
- Footnote
- Patient Education

The following endpoints are commonly used with Informed by MCG for Utilization Management.

- Alternatives
- Alternative Care Planning
- Alternative Levels of Care
- Benchmarks
- Care Planning
- Discharge Criteria
- Discharge Destination
- Discharge Planning
- Discharge Planning Tool
- Discharge Readiness
- Hospital Care Planning
- Evaluation Treatment

- Evidence Summary
- Extended Stay
- Hospital Care
- Indications
- Information Review
- Length of Stay
- Levels of Care
- Operative Status
- Operative Status Assignment
- Operative Status Criteria
- Preoperative Care Planning
- Prevention
- Recovery Course
- Reference Links
- Table of Contents

The following endpoints can be used to retrieve reference-related guideline information:

- Related codes

Other endpoints are available in the MCG Swagger web service test site that are not required for the Informed by MCG for Utilization Management use case.



In nearly all instances, *contentOwner* will always be GA.

4.2 Alternatives endpoint

`GET /api/v2/{contentOwner}/{edition}/{product}/Alternatives/{hsim}`

In API Version 2, the Alternatives endpoint retrieves the Alternatives section of a specific guideline. The following example shows the Alternatives to Admission section as it appears in the ISC guideline content.

Inpatient & Surgical Care > Optimal Recovery Guidelines > Cardiology > Heart Failure (M-190)

- Care Planning - Inpatient Admission and Alternatives
 - Clinical Indications for Admission to Inpatient Care
 - Alternatives to Admission
 - Alternative Care Planning

The following example shows how the response for the Alternatives endpoint typically displays.

```
{
  "Heading": "Alternatives to Admission",
  "Content": [
```



```
{
  "Text": "Alternatives include[[Cit:116500]][[Cit:199014]][[Cit:259522]][[Cit:203553]]
[[Cit:232674]][[Cit:2521781]][[Cit:121986]][[Cit:116354]][[Cit:259507]]:[[Ano:1]]",
  "Id": "226147694CA654FB",
  "PathId": "226147694CA654FB",
  "AnnotationLinks": [
    {
      "Id": "1",
      "AnnotationType": "ALT",
      "Link": "https://api-EndpointURL.com/api/v2/GA/27.0/ISC/0010445010b/annotation/1"
    }
  ]
}
...
```

4.3 Annotation endpoint

GET /api/{contentOwner}/{edition}/{product}/{hsim}/Annotation/{id}

The Annotation endpoint retrieves the content of an annotation. An annotation can contain supporting evidence, suggestions, and/or alternatives. In an Informed by MCG for Utilization Management web service response, an annotation contains a text description and a series of links to the citations that support the content of the annotation. Annotations are unique to each guideline; an annotation is assigned a different identifier for each guideline in which it appears.

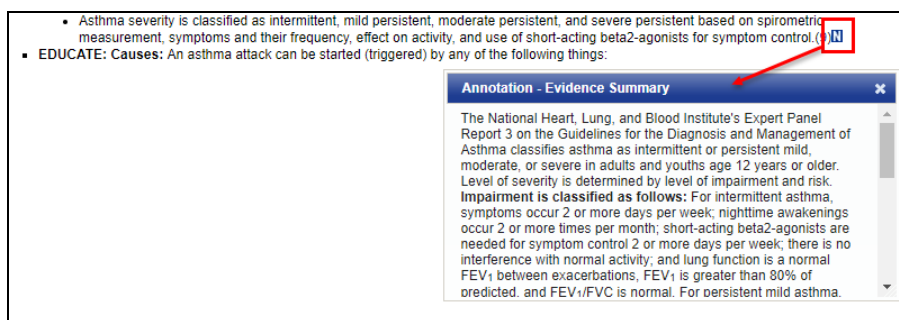
Annotations may include child citation elements.

4.3.1 About annotations

The MCG care guidelines are evidence based, and the annotations that accompany guideline text are an important part of each guideline. You must make all annotation content available to a user.

An annotation is a supporting reference that includes a set of citations. Annotations can exist in almost any of the guideline content.

You decide whether and how to include the reference information when you display guideline content to a user. In MCG applications, the reference appears as a link at the end of the applicable indication. If selected, the reference information appears in a separate window.



Annotations appear in the web service response under the AnnotationLinks element.

```
"AnnotationLinks": [
  {
    "Id": "1",
    "AnnotationType": "CRI",
    "Link": "https://api-EndpointURL.com/api/GA/27.0/ISC/0010445010b/annotation/1"
```

When you retrieve the content of an annotation using the Annotation endpoint, you receive the annotation text and the identifiers for any citations or references that support the annotation. The guideline HSIM is required to retrieve the text of an annotation.

MCG typically displays annotations in the indications content using the  symbol. Annotations also may be included in the Evidence Summary of a guideline.

Each annotation has an identifier and an associated type. These refer to the type of evidence that is being cited.

ID	Type	Evidence type
1	CRI	Criteria
2	ALT	Alternatives
3	LOS	Length of Stay
4	HOS	Hospitalization

4.4 Benchmarks endpoint

`GET /api/{contentOwner}/{edition}/{product}/Benchmarks/{hsim}`

Benchmark statistical data is available for some of the MCG guidelines. The benchmark statistical data provides MCG client organizations with a basis for measuring their performance and efficiency against a nationwide sample of actual patient cases.

The type of benchmark statistical data that is available varies depending on the care guideline product that you are using. For example, different statistical data is provided for Home Care guidelines than is for Inpatient & Surgical Care guidelines. The Content Guide for each content product describes, in more detail, the statistical data that is available and how it was calculated. To reach the Content Guide, open the guideline search page in CareWeb. To open the content product table of contents, select one of the guideline product links in the list of checkboxes at the top of the page. The Content Guide is one of the first entries.

When you retrieve data using the Benchmarks endpoint, you are retrieving the benchmark data for a single guideline.



The response includes data for both Commercial and Medicare patients. Before displaying the benchmark information to a user, you may want to filter out one of these options.

4.5 Bundle endpoint

`GET /api/bundle`

The Bundle endpoint retrieves a list of the available content bundles for the client ID and key provided by MCG.

A content bundle contains a set of guidelines that are specific to a particular use case or workflow. They are different from the MCG content solutions in that a content bundle may contain a subset of guidelines with a content solution, or it may contain guidelines and content from more than one MCG guideline solution.

This endpoint verifies whether the content solutions on the client are properly licensed. Only content solutions that the client can access appear in the response.



The Bundle endpoint may become deprecated in the future. MCG recommends using the [Product endpoint](#) instead.

4.6 Care Planning endpoints

```
GET /api/v2/{contentOwner}/{edition}/{product}/CarePlanning/{hsim}
GET /api/v2/{contentOwner}/{edition}/{product}/HospitalCarePlanning/{hsim}
GET /api/v2/{contentOwner}/{edition}/{product}/AlternativeCare/{hsim}
GET /api/v2/{contentOwner}/{edition}/{product}/PreoperativeCarePlanning/{hsim}
```

In API Version 2, the CarePlanning endpoints retrieve the following Behavioral Health Guideline (BHG) sections:

- Care Planning and Evaluation
- Hospital Care
- Alternative Care Planning
- Preoperative Care Planning

The following example shows the Care Planning and Evaluation section as it appears in the BHG content.

Behavioral Health Care > Behavioral Health Level of Care Guidelines > Outpatient Behavioral Health Level of Care, Adult (B-901-AOP)

Outpatient Behavioral Health Level of Care, Adult

ORG: B-901-AOP (BHG)

[Print View](#)

- Admission Guidelines
- Recovery Course
- **Care Planning and Evaluation**
- Discharge Guidelines
- Discharge Planning
- References
- Footnotes

The following example shows how the response for the CarePlanning endpoint typically displays.

```
{
  "Heading": "Care Planning and Evaluation",
  "Content": [
    {
      "Text": "Planning and evaluating appropriate care should address[[Cit:211522]][[Cit:160747]]",
      "Id": "1ABD026E52412E4F",
      "PathId": "1ABD026E52412E4F",
    }
  ]
}
```

```
"CitationLinks": [
  {
    "Id": "211522",
    "DisplayValue": "1",
    "Link": "https://api-EndpointURL.com/api/v2/GA/27.0/citation/211522"
  },
  {
    ...
  }
]
```

4.7 Citation endpoint

GET /api/{contentOwner}/{edition}/Citation/{id}

The Citation endpoint retrieves the text of a citation.

4.7.1 About citations

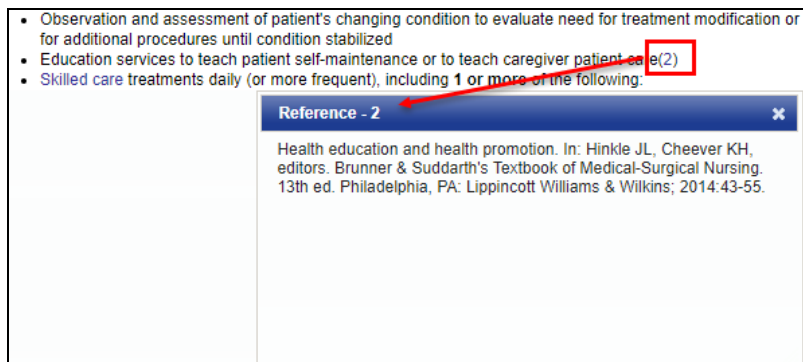
The care guidelines are evidence based. An important part of the guideline content is to identify the sources of information used to build the guideline. These are added to the guideline content as *citations*. The citations that accompany guideline text are an important part of each guideline. You must make all citation content available to a user.

Citations are defined once in a content product and used globally across all guidelines in that content product. Each citation is assigned a unique identifier that remains the same regardless of in which guideline the citation appears.

A citation is a reference to supporting documentation and information. One or more citations can be associated with an indication. Citations appear in nearly all of the guideline content. The text of citations is included in the XML response immediately following the element in which they are referenced.

You decide whether and how to include the reference information when presenting guideline content to a user. In MCG applications, the reference appears as a link at the end of the applicable indication. If selected, the reference information appears in a separate window.

The following example shows how MCG typically displays a citation within a guideline.



Citations are defined once in a content product and used globally across all guidelines in that content product. Each citation is assigned a unique identifier that remains the same regardless of which guideline the citation appears.

Citations appear in the response as CitationLinks.

```
"CitationLinks": [
  {
    "Id": "7128",
    "DisplayValue": "1",
    "Link": "https://api-EndpointURL.com/api/GA/27.0/citation/7128"
```

A citation entry may include a unique citation identifier used to retrieve the citation content. The citation identifier is independent of any guideline in which the citation appears. You can look up the content of a citation directly using the identifier; no guideline HSIM code or other identifier is required.

There is a display value for each of the citation entries, shown in red in the preceding example. You can append this value to the end of the indication if you choose to include reference materials with the indication content.

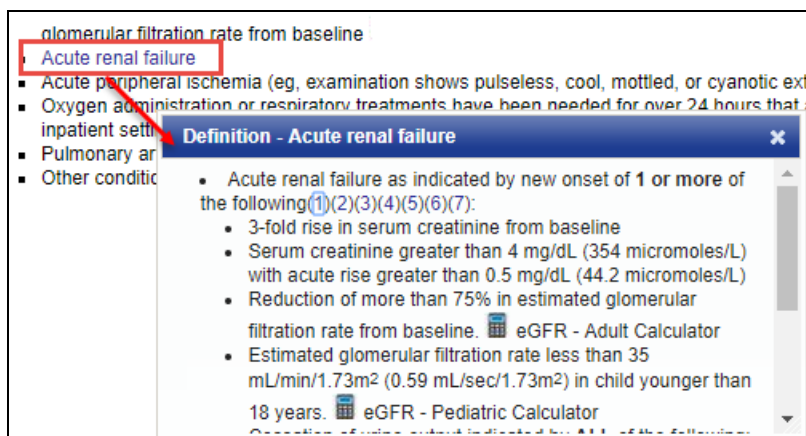
4.8 Definition endpoint

GET /api/{contentOwner}/{edition}/Definition/{id}

The Definition endpoint retrieves the content of a definition.

4.8.1 About definitions

A definition provides additional detailed information about a term used in the guideline content. Definitions use the same format and structure as an indications section; specifically, one or more parent indications with child indications and logic rules determine whether the parent indication has been met. Definitions can appear as read only, or they can appear interactively as a worksheet to help a user determine if the defined term is applicable to a patient. For more information, see the [Indication endpoints](#).



Definitions are defined once within an MCG content product and used globally across all guidelines within that content product. A single definition will always be assigned the same identifier regardless of in which guideline it appears.

4.8.1.1 Definition text

Definition text commonly contains additional definitions. Whenever you retrieve the text of a definition, check to see if it includes other definitions and retrieve the content for those as well.

The text of a definition frequently includes references to another definition, annotation, footnote, citation, link to another guideline, and calculators. Be prepared to retrieve the text for all of these elements each time you make definition content available to a user.

4.8.1.2 Content of a definition

In some instances of the Informed by MCG for Utilization Management web service, the content of a definition is automatically embedded in the response when you retrieve guideline data. If not, to retrieve the content of the definition, use the content edition and definition.

4.8.1.3 Definition reference link

The following example shows how the reference link (the **Link** element) to the corresponding definition text appears in the response.

```
[,
  "ContentItems": [
    {
      "Text": "Tachycardia that persists despite appropriate treatment (eg, volume repletion,
treatment of pain, treatment of underlying cause)",
      "Id": "613DBFDF98ED2E6C",
      "PathId": "160FCF09404835C5-B411A0BF7724D587-613DBFDF98ED2E6C",
      "Definitions": [
        {
          "Id": "DEF0081",
          "Title": "Tachycardia",
          "Link": "https://api-EndpointURL.com/api/GA/27.0/definition/DEF0081",
          "ContentItems": [
            {
```

4.9 Discharge endpoints

With the Discharge endpoints, you can perform the following actions:

- Retrieve the content of the Discharge Planning Tool (a utility/worksheet) provided by MCG with each edition of the care guidelines.
- Retrieve the Discharge Destination content of a specific guideline.
- Retrieve the Discharge Planning, Discharge Criteria, and Discharge Readiness sections of the guideline content.

Here are more details.

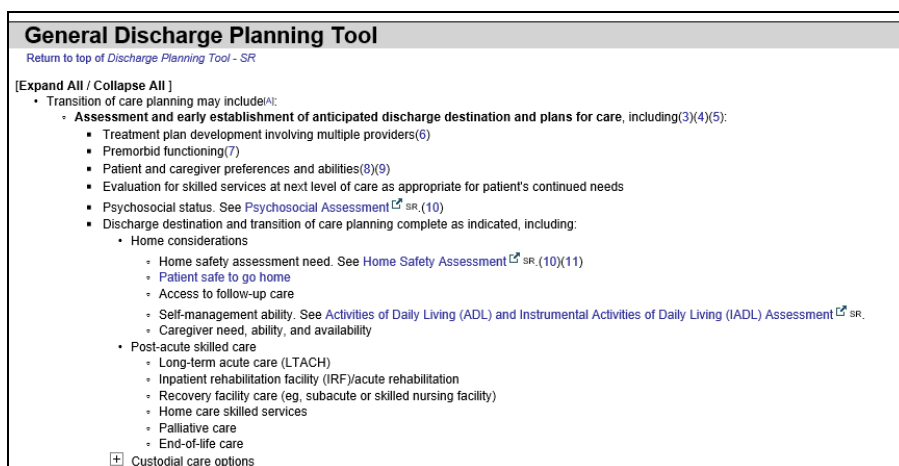
4.9.1 DischargePlanningTool endpoint

GET /api/{contentOwner}/{edition}/DischargePlanningTool

The DischargePlanningTool response provides a list of selectable indications, each of which can contain definitions, citations, annotations, footnotes, and links to other guidelines. Unlike other indication sections in the care guidelines, it does not include logic rules and therefore cannot be used to automatically calculate whether a patient is ready for discharge. For more information, see the [Indication endpoints](#).

There is just one Discharge Planning Tool for each care guideline *edition*. Using this utility, the Discharge Planning Tool content you retrieve will not vary depending on which MCG content products and/or bundles you are using as long as they are all part of the same edition.

The following image is an example of part of the Discharge Planning Tool.



4.9.2 DischargeDestination endpoint

`GET /api/{contentOwner}/{edition}/{product}/DischargeDestination/{hsim}`

The DischargeDestination endpoint retrieves the Discharge Destination content of a specific guideline. This endpoint can have two different DischargeDestination response structures as shown.

Example: Dual heading response structure specific to certain BHG guidelines

```
{
  "Heading": "Discharge Destination",
  "Sections": [
    {
      "Heading": "Usual",
      "Content": [
        {
          "Text": "Outpatient care. See Other Psychiatric Disorders: Outpatient Care.",
          "Id": "C4EA4498F5EE64CA",
          "PathId": "C4EA4498F5EE64CA",
          "GuidelineLinks": [
            {
              "ContentVersion": "27.0",
              "Title": "Other Psychiatric Disorders: Outpatient Care",
              "Hsim": "00215437002",
            }
          ]
        }
      ]
    }
  ]
}
```

```

        "Product": "BHG"
    }
}
{
    "Heading": "Alternate",
    "Content": [
        {
            "Text": "Intensive outpatient program. See Other Psychiatric Disorders:
Intensive Outpatient Program.",
            "Id": "A5FE3954C054697F",
            "PathId": "A5FE3954C054697F",
            "GuidelineLinks": [
                {
                    "ContentVersion": "27.0",
                    "Title": "Other Psychiatric Disorders: Intensive Outpatient Program",
                    "Hsim": "00215436904",
                    "Product": "BHG"
                }
            ]
        }
    ]
}

```

Example: Single heading response structure for all other guidelines

```

{
    "Heading": "Discharge Destination",
    "Content": [
        {
            "Text": "Post-hospital levels of admission may include:",
            "Id": "6F8AF10574E925A5",
            "PathId": "6F8AF10574E925A5",
            "ContentItems": [
                {
                    "Text": "Home.",
                    "Id": "D85D7050F0539057",
                    "PathId": "6F8AF10574E925A5-D85D7050F0539057"
                },
                {
                    "Text": "Home healthcare. See Home Care Indications for Admission Section in
Asthma guideline in Home Care.",
                    "Id": "D56F1B263208B5EB",
                    "PathId": "6F8AF10574E925A5-D56F1B263208B5EB",
                    "GuidelineLinks": [
                        {
                            "ContentVersion": "27.0",
                            "Title": "Home Care Indications for Admission Section",
                            "Hsim": "0090517100f",
                            "Product": "HC"
                        }
                    ]
                }
            ]
        },
        {
            "Text": "Recovery facility care. See Recovery Facility Care Indications for
Admission Section in Asthma guideline in Recovery Facility Care.",
            "Id": "17C6C21554074AED",
            "PathId": "6F8AF10574E925A5-17C6C21554074AED",
            "GuidelineLinks": [
                {
                    "ContentVersion": "27.0",
                    "Title": "Recovery Facility Care Indications for Admission Section",
                    "Hsim": "0040517100d",
                }
            ]
        }
    ]
}

```



```

        "Product": "RFC"
      }
    ]
  }
}
],
"SectionPath": "/Discharge/Discharge_Destination",
"CustomGuideline": false
}

```

4.9.3 DischargePlanning endpoint

`GET /api/v2/{contentOwner}/{edition}/{product}/DischargePlanning/{hsim}`

In API Version 2, the DischargePlanning endpoint retrieves the Discharge Planning section of a specific guideline. The following example shows the Discharge Planning section as it appears in the ISC guideline content.

Inpatient & Surgical Care > Optimal Recovery Guidelines > Cardiology > Heart Failure (M-190)

Discharge Planning

[Return to top of Heart Failure - ISC](#)

[Expand All / Collapse All]

- Discharge planning includes^[0]:
 - Assessment of needs and planning for care, including⁽⁹³⁾:
 - Develop treatment plan (involving multiple providers as needed).
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address social determinants of health (eg, housing, food).

The following example shows how the response structure for the DischargePlanning endpoint typically displays.

```

{
  "Heading": "Discharge Planning",
  "Content": [
    {
      "Text": "Discharge planning includes[0]:",
      "Id": "8952F915A6667D85",
      "PathId": "8952F915A6667D85",
      "FootnoteLinks": [
        {

```

...

4.9.4 DischargeCriteria endpoints

`GET /api/v2/{contentOwner}/{edition}/{product}/DischargeCriteria/{hsim}`

`POST /api/v2/{contentOwner}/{edition}/{product}/DischargeCriteria/{hsim}`

In API Version 2, the GET DischargeCriteria endpoint retrieves the Discharge Criteria section of a specific guideline. The POST DischargeCriteria endpoint calculates whether the discharge criteria have been met. The input to this function is a list of discharge criteria that a user has selected.

MCG calculates the discharge criteria of Met status based on the guideline requirements and user selections.

4.9.5 DischargeReadiness endpoint

`GET /api/v2/{contentOwner}/{edition}/{product}/DischargeReadiness/{hsim}`

In API Version 2, the GET DischargeReadiness endpoint retrieves the Discharge Readiness section of a specific guideline.

4.10 Extended Stay endpoint

`GET /api/{contentOwner}/{edition}/{product}/ExtendedStay/{hsim}`

The ExtendedStay endpoint retrieves the entire content of the Extended Stay section or Extended Visit section of a specific guideline. This information is view-only; there is no corresponding POST endpoint to calculate met or not met status. The structure of the content response is like the Indications response. For more information, see the [Indication endpoints](#).

4.10.1 About extended stay/visit

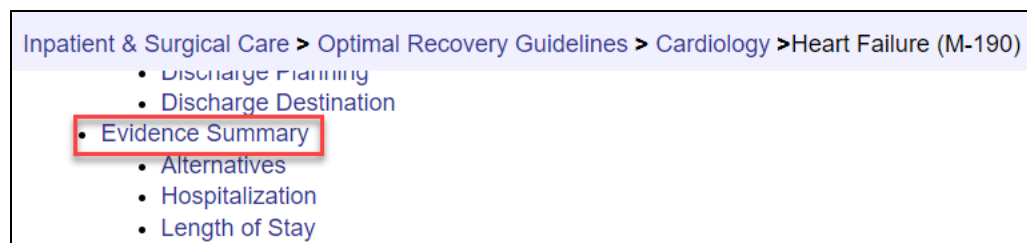
Some of the care guidelines include an Extended Stay section or Extended Visit section. If present in a guideline, this section describes the circumstances under which the expected length of inpatient stay, or the expected number of outpatient care visits might exceed the standard guideline recommendations.

The content of this section consists of a series of parent/child indications. The entries in an Extended Stay or Extended Visit can contain footnotes, annotations, citations, and definitions.

4.11 Evidence Summary endpoint

`GET /api/v2/{contentOwner}/{edition}/{product}/EvidenceSummary/{hsim}`

In API Version 2, the Evidence Summary endpoint retrieves the Evidence Summary section of a specific guideline. The following example shows the Evidence Summary section as it appears in the ISC guideline content.



The following example shows how the response for the Evidence Summary endpoint typically displays.

```
{
  "Heading": "Evidence Summary",
}
```

```
"Sections": [
  {
    "Heading": "Alternatives",
    "Paragraphs": [
      "Analysis of 327 patients with acute heart failure treated in an observation unit found that
73% of patients ...]
    },
  ],
  ...
]
```

4.11.1 Supplemental Medicare criteria for medical guidelines

To meet CMS requirements, the Rationale and Related CMS Guidance supplemental Medicare criteria subsections have been added to the evidence summary for medical ORG and GRG guidelines. This criteria addresses the two-midnight rule and is specifically intended for use with Medicare Advantage and Medicare fee for service patients.

The following example shows how the subsections appear in ISC content.

Evidence Summary

[Return to top of Gastroenteritis - ISC](#)

Criteria

[Return to top of Gastroenteritis - ISC](#)

The evidence for the clinical indications found in this guideline includes 2 published peer reviewed articles, 2 specialty society or other evidence-t

Alternatives

[Return to top of Gastroenteritis - ISC](#)

Outpatient management with oral replacement therapy can be accomplished in the majority of patients with gastroenteritis.(1)(2)(4)(10) (**EG 2**) Or (11) (**EG 2**) An international study of 3702 adults and children with culture-confirmed *Campylobacter* infection reports 91% were treated as outpat analysis of 2736 adults and children with culture-confirmed non-typhoidal *Salmonella* infection reports 80% were treated as outpatients; however,

Length of Stay

[Return to top of Gastroenteritis - ISC](#)

Analysis of national hospital discharge data shows 39% of hospitalized adult patients with the principal diagnosis of gastroenteritis discharged in ;

Rationale

[Return to top of Gastroenteritis - ISC](#)

Medical MCG care guidelines help the clinician to identify patient-specific complex clinical factors that make it reasonable to expect a necessity fo patient to inpatient care. MCG care guidelines also support the clinician in the decision to appropriately admit a patient to inpatient care when thei

Related CMS Coverage Guidance

[Return to top of Gastroenteritis - ISC](#)

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinat

Code of Federal Regulations (CFR): 42 CFR 412.3(8); 42 CFR 419.22(n)(61); 42 CFR 422.101(62)

The following example shows how the Evidence Summary endpoint response typically displays.

```
]
  },
  {
    "Heading": "Rationale",
    "Paragraphs": [
      "Medical MCG care guidelines help the clinician to identify patient-specific complex clinical
factors ...]
    },
  ],
  {
    "Heading": "Related CMS Coverage Guidance",
    "Paragraphs": [

```

```
"This guideline supplements but does not replace, modify, or supersede existing Medicare
regulations ...]
},
...
```

4.12 Footnote endpoints

A footnote in the guideline content usually provides supporting information or detail. MCG typically displays a footnote as a letter enclosed by brackets.

Footnotes are identified relative to a guideline or a definition, which is why there are two Footnote endpoints.

- If a footnote is used anywhere in a guideline except a definition, you must retrieve it using the guideline ID (HSIM code) and footnote ID.
- If a footnote is used within a definition, you must retrieve it based on both the definition ID and footnote ID.

Footnotes may include child citation elements.

4.12.1 Footnote endpoint (using the footnote ID)

```
GET /api/{contentOwner}/{edition}/{product}/{hsim}/Footnote/{id}
```

This endpoint retrieves the text of a footnote using a *guideline ID* (HSIM code) and footnote ID.

4.12.2 Footnote endpoint (using the definition ID)

```
GET /api/{contentOwner}/{edition}/Definition/{definitionId}/Footnote/{id}
```

This endpoint retrieves the text of a footnote using a *definition ID* and footnote ID.

4.12.3 About footnotes

The MCG care guidelines are evidence based, and the footnotes that accompany guideline text are an important part of each guideline. Footnotes contain valuable information for users, such as supplemental clinical information. You must make all footnote content available to a user.

A footnote is a reference to supporting documentation and information. One or more footnotes can be associated with an indication. Footnotes can exist in almost any of the guideline content. The text of footnotes is included in the XML response immediately following the element in which they are referenced.

When you display guideline content to your user, you must decide whether and how to include the reference information. In MCG applications, the reference appears as a link at the end of the applicable indication. If selected, the reference information appears in a separate window, as shown.

- There are no acute hospital care needs.^[A]
- The patient has *Intense and complex* care needs that make recovery facility care safer and more practical than hospital care at a lower level.

Footnote - A

An observational study of 318 patients admitted to subacute care facilities after hospitalization found improvements in blood glucose, HbA1c, BNP, and CRP and significant improvements in functional status scores. No improvements were seen in serum creatinine and decline in hemoglobin levels, particularly in patients that had hospital readmissions within 6 months of subacute care discharge.⁽¹⁾

The following example shows a footnote within a definition.

- Serum calcium greater than 14 mg/dL (3.5 mmol/L)⁽²⁾
- Serum calcium greater than 12 mg/dL (3.0 mmol/L) with *Altered mental status*, severe volume depletion requiring volume IV fluid resuscitation, *Hypotension*, significant arrhythmia, heart block, or digitalis toxicity⁽²⁾
- Serum phosphorus less than 1 mg/dL (0.32 mmol/L)
- Toxic drug level or poisoning causing or likely to cause neurologic abnormalities or hemodynamic instability
- Less severe laboratory abnormalities

Definition - Hypotension

- Hypotension as indicated by **ALL** of the following⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾:
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure as indicated by **1 or more** of the following:
 - New onset of SBP less than 90 mm Hg in adult or child 10 years or older
 - New decrease in SBP greater than 40 mm Hg in adult or child 10 years or older
 - Mean arterial pressure ^[A] less than 70 mm Hg in adult or child 10 years or older
 - New onset of SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age

Footnotes appear under the FootnoteLinks element in the response. Each footnote entry includes the unique identifier used to retrieve the footnote content.

```

"FootnoteLinks": [
  {
    "Id": "f0428100b_305_1_48",
    "Link": "https://api-EndpointURL.com/api/GA/28.0/ISC/0010428100b/footnote/f0428100b_305_1_48",
    "DisplayValue": "[A]"
  }
]

```

There is a display value for each of the footnote entries, shown in red in the preceding example. This is the value you can append to the end of the indication if you choose to include reference materials with the indication content.

4.13 Guideline endpoints

You can retrieve a list of guidelines for a specific product or content bundle.

The Guideline endpoints retrieve only the latest edition. Minor versions are available through other endpoints.

4.13.1 Guidelines endpoint

`GET /api/{contentOwner}/{edition}/{product}/guidelines`

Identifies and lists the guidelines that are contained in a content bundle.

It is possible to concurrently license more than one edition/version of guideline content. This endpoint retrieves only the latest edition. Minor versions are available through other endpoints.

4.13.2 GuidelineContent endpoint

`GET /api/{contentOwner}/{edition}/{product}/guidelineContent`

Identifies and provides links to the resources available for each guideline within a product or content bundle. The types of resources vary depending on the content product or content bundle.

Examples:

- Assessment section
- Plan of Care section in the Chronic Care assessment
- Transitions of Care assessment

4.14 Hospital Care endpoint

`GET /api/v2{contentOwner}/{edition}/{product}/HospitalCare/{hsim}`

In API Version 2, the HospitalCare endpoint retrieves the Hospital Care section of a specific guideline including the clinical indication logic. The following example shows the Hospital Care section as it appears in the ISC guideline content.

Inpatient & Surgical Care > Common Complications and Conditions > Alcohol and Psychoactive Substance Withdrawal: Common Complications and Conditions (CCC-035)

Alcohol and Psychoactive Substance Withdrawal: Common Complications and Conditions

CCC: CCC-035 (ISC)
[Link to Codes](#)

- Prevention
- Clinical Indications for Inpatient Care
- Alternatives to Inpatient Care
- **Hospital Care**
- Discharge
- References
- Footnotes
- Codes

The following example shows how the response for the HospitalCare endpoint typically displays.

```
{
  "Heading": "Hospital Care",
  "Paragraph": "",
  "Content": [
    {
      "Text": "Evaluation may include[[Cit:164241]][[Cit:228084]][[Cit:2515856]][[Cit:2508440]]",
      "Id": "DC9B0D83093234E3",
      "PathId": "DC9B0D83093234E3",
    }
  ]
}
```

```
"CitationLinks": [
  {
    "Id": "164241",
    "DisplayValue": "2",
    "Link": "https://api-EndpointURL.com/api/v2/GA/27.0/citation/164241"
  },
  {
    ...
  }
]
```

4.15 Indication endpoints

There are two Indication endpoints: GET Indications and POST Indications. Use these endpoints to calculate whether an admission or procedure is warranted according to the care guidelines.

To calculate the indications of Met/Not Met status

1. Retrieve the content of the indications section in question using the GET Indication endpoint.
2. Present the indications to a user and allow them to select the indications that are present in the patient case.
3. Submit the selected responses using the POST Indication endpoint. The status of MET or Not Met is returned in the endpoint response.

4.15.1 GET Indication endpoints

```
GET /api/{contentOwner}/{edition}/{product}/Indications/{hsim}
GET /api/{contentOwner}/{edition}/{product}/AuthorizationIndications/{hsim}
GET /api/v2/{contentOwner}/{edition}/{product}/QualityMeasures/{hsim}
GET /api/{contentOwner}/{edition}/{product}/OperativeStatusCriteria/{hsim}
GET /api/{contentOwner}/{edition}/{product}/OperativeStatusAssignment/{hsim}
```

The GET Indication endpoints let you retrieve the contents of a specific guideline.

In Version 2 of the Indication endpoints, the QualityMeasures endpoint retrieves the Quality Measures section for various assessments and guidelines.

4.15.2 POST Indication endpoints

```
POST /api/{contentOwner}/{edition}/{product}/Indications/{hsim}
POST /api/{contentOwner}/{edition}/{product}/AuthorizationIndications/{hsim}
```

Use these endpoints to calculate whether the indications for a particular guideline have been met. The input to this function is a list of indications that a user has selected. MCG calculates the indication of Met status based on the guideline requirements and user selections.

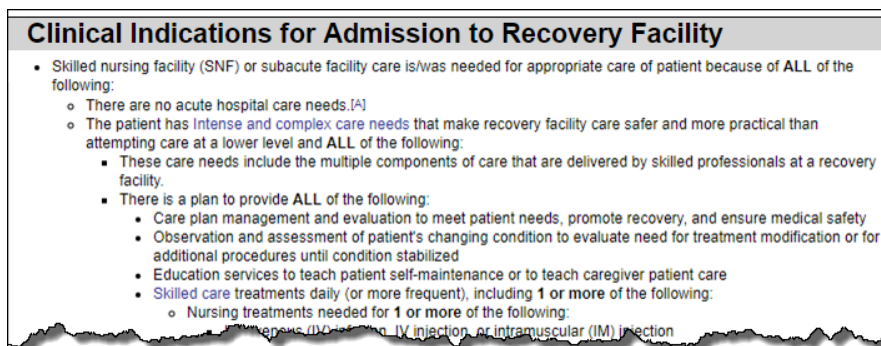


To activate and use the AuthorizationIndications endpoint in your preauthorization workflows, contact your MCG Account Executive for assistance.

4.15.3 About indications

Most of the criteria and assessment content in the care guidelines is contained in indication sections. Each indication section lists the indications and rules used to determine whether the patient meets the guideline criteria for a particular course of action.

The following example shows a typical Indications guideline section as it appears in the guideline content.



Different combinations of indications may determine whether the referral, admission, or procedure is warranted. For example, in the preceding Angina guideline, admission to inpatient care is considered warranted in two circumstances:

- The patient has both unstable angina and any symptoms that warrant acute intervention, as defined by the list of sub-indications.
- There is any evidence that the patient has had an MI.

4.15.3.1 Components of an indication

An indication contains a series of components:

- The indication identifier.
- The path identifier.
- A Selected element that indicates whether a user has selected the indication (for POST endpoints).
- If there are sub-indications, logic (the ListLogic element) to determine how many of the child indications must be met for the parent indication to be met.
- A Status element which indicates whether the indication requirements have been met (for POST endpoints).
- The text description of the indication.
- Definitions which provide additional detail regarding a term used in the indication. For example, there is a definition for Hemodynamic Instability that assists a user in determining whether or not hemodynamic instability is present.
- Links to supporting reference information, such as footnotes, annotations, and citations.
- Links to other guidelines and guideline components.
- Appropriate indications are tagged with the Quality Measures [QM] designation.
- A NoteRequired attribute for supplemental Medicare criteria-related indications.

4.15.3.2 Identifiers

Every indication has two identifiers:

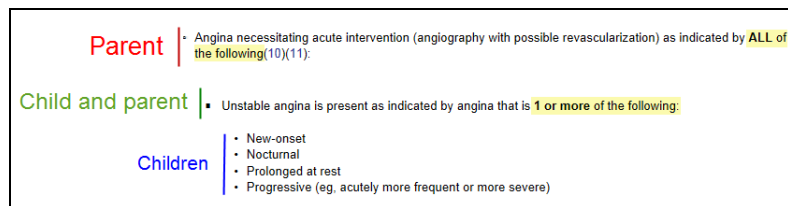
- Indication identifier (Id): For identifying "1 or more of the following" procedures within a care guideline.
- Path identifier (PathId): For identifying instances of an indication's path within a care guideline.

These identifiers are included in the initial response when you request the content of a guideline from the Informed by MCG for Utilization Management web service. The path identifier combined with the indication identifier can be used to identify one or more instances of an indication within a care guideline.

```
{
  "Heading": "Inpatient Care",
  "Content": [
    {
      "Text": "Admission is indicated by 1 or more of the following:",
      "ListLogic": "1 or more of the following",
      "Id": "D40F5C2F1BA84A8A",
      "PathId": "D40F5C2F1BA84A8A",
      "ContentItems": [
        {
          "Text": "Angina necessitating acute intervention (angiography with possible
revascularization) as indicated by ALL of the following:",
          "ListLogic": "All of the following",
          "Id": "D9C830E7010C4D34",
          "PathId": "D40F5C2F1BA84A8A-D9C830E7010C4D34",
          "ContentItems": [
            {
              "Text": "Unstable angina is present due to ischemic symptoms (eg, chest pain,
dyspnea) and 1 or more of the following:",
              "ListLogic": "1 or more of the following",
              "Id": "44F869C5AD9227F9",
              "PathId": "D40F5C2F1BA84A8A-D9C830E7010C4D34-44F869C5AD9227F9",
              "ContentItems": [
                {
                  "Text": "New-onset symptoms",
                  "Id": "EEE25A4A807EBB46",
                  "PathId": "D40F5C2F1BA84A8A-D9C830E7010C4D34-44F869C5AD9227F9-EEE25A4A807EBB46"
                },
                ...
              ]
            }
          ]
        }
      ]
    }
  ]
}
```

4.15.3.3 Child indication

An indication can contain a series of sub-indications. These are referred to as parent and child indications. A child indication can, in turn, be a parent indication to other child indications. There are no restrictions on the number of parent/child sub-levels within an indication.



4.15.3.4 Selected

There is a Selected element for each of the indications, whether parent or child. MCG calculates and sets the value of the Selected element based on the indications submitted through the POST request.

```
{
  "Heading": "Clinical Indications for Admission to Inpatient Care",
  "Content": [
    {
      "Status": "Met",
      "Selected": true,
      "Text": "Admission to Inpatient Level of Care for Adult (LOCUS Level 6 - Medically Managed Residential Services, Composite Score 28 or more) is indicated due to ALL of the following[[Foot:fDEF_0955_0001]][[Foot:fDEF_0955_0002]][[Foot:fDEF_0955_0003]][[Foot:fDEF_0955_0004]]:",
      "ListLogic": "All of the following",
      "Id": "E6E22B871EB3C246",
      "PathId": "E6E22B871EB3C246",
      "FootnoteLinks": [
        {
          "Id": "fDEF_0955_0001",
          "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/BHG/0021543620b/footnote/fDEF_0955_0001",
          "DisplayValue": "[A]"
        }
      ],
      ...
    }
  ]
}
```

4.15.3.5 ListLogic

The ListLogic element specifies how many of the child indications must be selected in order for that parent indication to be met.

The following example illustrates how a ListLogic element and its child indications appear in the response.

```
{
  "Heading": "Inpatient Care",
  "Content": [
    {
      "Text": "Admission is indicated by 1 or more of the following:",
      "ListLogic": "1 or more of the following",
      "Id": "D40F5C2F1BA84A8A",
      "PathId": "D40F5C2F1BA84A8A",
      "ContentItems": [
        {
          "Text": "Angina necessitating acute intervention (angiography with possible revascularization) as indicated by ALL of the following:",
          "ListLogic": "All of the following",
          "Id": "D9C830E7010C4D34",
          "PathId": "D40F5C2F1BA84A8A-D9C830E7010C4D34",
          "ContentItems": [
            {
              "Text": "Unstable angina is present due to ischemic symptoms (eg, chest pain, dyspnea) and 1 or more of the following:",
              "ListLogic": "1 or more of the following",
              "Id": "44F869C5AD9227F9",
              "PathId": "D40F5C2F1BA84A8A-D9C830E7010C4D34-44F869C5AD9227F9",
              "ContentItems": [
                {
                  ...
                }
              ]
            }
          ]
        }
      ]
    }
  ]
}
```

```
"Text": "New-onset symptoms",
"Id": "EEE25A4A807EBB46",
"PathId": "D40F5C2F1BA84A8A-D9C830E7010C4D34-44F869C5AD9227F9-EEE25A4A807EBB46"
},
```

...

The ListLogic value is contained in the ListLogic element of an indication, and is usually repeated in text form within the indication description.

In the preceding example, you must select only one of the child indications for the parent indication to be true. Use the following ListLogic options to determine Met/Not Met status:

- All of the following
- 1 or more of the following
- 2 or more of the following
- 3 or more of the following
- 4 or more of the following
- 5 or more of the following

In addition, there are ListLogic types that do not trigger a Met/Not Met status:

- C Current Role Remains Uncertain
- E Examples include
- N Note

Every indication, whether parent or child, has an identifier and can be submitted to MCG as selected. This can override the logic rule for how many of the child indications must be selected for the parent indication to be true. You can decide whether to allow a user to select a parent indication directly, or to limit a user to selecting only the child indications and have MCG determine whether the parent indication has been met.



Even though the ListLogic information is included in the GET response, you must post the selected indications to get a final Met or Not Met decision.

4.15.3.6 Status

The POST Indications endpoint includes a Status element for each indication that shows whether the indication has been met. MCG calculates and sets the value of the Status element.

The highest level Status element in the response indicates whether the patient has met or not met the overall requirements for a particular course of action, such as for an inpatient admit or referral to another provider. Subsequent Status elements reflect the status of each indication that contributed to the overall Met/Not Met status.

```
{
  "Heading": "Clinical Indications for Admission to Inpatient Care",
  "Content": [
    {
      "Status": "Met",
      "Text": "Admission is indicated for 1 or more of the following[[Cit:203741]][[Cit:188728]]
```

```
[[Cit:189013]][[Cit:203502]]:",
  "ListLogic": "1 or more of the following",
  "Id": "8DE994F59626DF23",
  "PathId": "8DE994F59626DF23",
  "CitationLinks": [
    ...
  ],
  "Sections": null,
  "SectionPath": "/Care_Planning_-_Inpatient_Admission_and_Alternatives/Clinical_Indications_for_
Admission_to_Inpatient_Care",
  "Status": "Met",
  "CustomGuideline": false,
  "CustomGuidelineDisclaimer": ""
}
```

4.15.3.7 Text

The Text element contains a text description of the indication. Additionally, it may contain a reference to another guideline or a guideline tool or calculator.

4.15.3.8 Links to other guidelines and guideline tools

Links to other guidelines may appear in the indication text, such as "See Myocardial Infarction guideline." These links are inactive in the response.

```
{
  "ContentItems": [
    {
      "Status": "Not Met",
      "Text": "Clinically active diabetes, see Angina with Clinically Active Diabetes
MCM",
      "Id": "7DD7D7C861C03DD6",
      "Definitions": [
        {
          "Id": "DEF_03126",
          "Title": "Clinically active diabetes",
          "ContentItems": [
            {
              "Text": "Clinically active diabetes refers to diabetes necessitating
treatment while hospitalized with insulin.",
              "Id": "831691C4C8C619AB"
            }
          ]
        }
      ]
    }
  ]
}
```

4.15.3.9 Calculators

Some of the indications reference a calculator. These are typically a worksheet that a user can use to calculate a score or measurement.

Calculators are not currently active in the Informed by MCG for Utilization Management web service. The references in the response are placeholders only.

```
{
  "Text": "TIMI risk score of 2 or greater[[Cit:184563]]TIMI Calculator",
  "Id": "22E33B3BC43C4416",
  "CitationLinks": [
    {
      "Id": "184563",
      "DisplayValue": "10",
      "Link": "https://api-EndpointURL.com/api/GA/28.0/citation/184563"
    }
  ],
  "CalculatorTypes": [
    "TIMI"
  ]
}
```

4.15.3.10 Quality Measures

In API Version 2, appropriate indications have been tagged with the QM designation in the medical necessity guidelines (Home Care). The GET QualityMeasures endpoint lets you retrieve the associated text, which describes the intent of the QM tag and how information from questions may be requested by various accrediting organizations.

Quality measures are intended to highlight areas of focus for major national quality initiatives. They are presented for reference purposes only and are neither an assurance of compliance nor a prescribed measurement set. The selection of accreditation organization, quality initiatives, measurements, and data collection instruments are at the user's discretion.

[Home Care](#) > [Optimal Recovery Guidelines](#) > [Cardiology](#) > [Angina \(M-2040\)](#)

Quality Measures

[Return to top of Angina - HC](#)

- NCQA HEDIS Measures include(67):
 - Patient medication reconciliation is completed.
 - The patient's transition of care needs are assessed and addressed.
 - The patient's psychosocial status (eg, depression screening) is assessed and addressed.
 - The patient's risk for falls is assessed and addressed.
 - The patient's smoking and tobacco use is assessed and addressed.
 - The patient's readmission risk is assessed and addressed.
- CMS Meaningful Measures include(68):

MCG displays a QM tag as two letters enclosed by brackets **[QM]**. You must retrieve it using the guideline ID (HSIM code).

Clinical Indications for Admission to Home Healthcare

[Return to top of Angina - HC](#)

- Home healthcare is/was needed for appropriate care of the patient because of **ALL** of the following[A][B][C](1)(2)(3)(4):
 - The patient is **Homebound**.
 - The patient's needs can be safely met with **Intermittent home care**.
 - The patient needs education on condition and self-management care with teach-back **QM**(5)(6)
 - **Skilled care** at home is required for **1 or more** of the following:
 - Cardiovascular status assessment(7)
 - Clinician assessment after emergency department or observation care without hospital admission(8)(9)

The following example comes from the QualityMeasures endpoint for Angina - HC. It shows where the QM tag placeholder is located at the end of the question text like a citation, annotation, or footnote.

Example: QM tag

```
...
{
  "Text": "The patient needs education on condition and self-management care with teach-back.
  [[QM]][[Cit:196514]][[Cit:206060]][[Cit:2503145]]",
  "Id": "E621BBB514AC0AE7",
  "PathId": "3F5565558196D0A1-E621BBB514AC0AE7",
  "CitationLinks": [
    {
      "Id": "196514",
      "DisplayValue": "2",
      "Link": "https://api.EndpointURL.com/api/v2/GA/27.0/citation/196514"
    },
    {
      "Id": "206060",
      "DisplayValue": "3",
      "Link": "https://api.EndpointURL.com/api/v2/GA/27.0/citation/206060"
    },
    {
      "Id": "2503145",
      "DisplayValue": "4",
      "Link": "https://api.EndpointURL.com/api/v2/GA/27.0/citation/2503145"
    }
  ]
},
...

```

Example: Angina - HC output

```
{
  "Content": [
    {
      "Text": "NCQA HEDIS Measures include[[Cit:206732]]:",
      "Id": "E59072E420E51506",
      "PathId": "E59072E420E51506",
      "CitationLinks": [
        {
          "Id": "206732",
          "DisplayValue": "74",
          "Link": "https://api.EndpointURL.com/api/v2/GA/27.0/citation/206732"
        }
      ]
    },
    "ContentItems": [

```

```
{
  "Text": "Patient medication reconciliation is completed.",
  "Id": "29A49FD0A74D9754",
  "PathId": "E59072E420E51506-29A49FD0A74D9754"
},
{
  "Text": "The patient's transition of care needs are assessed and addressed.",
  "Id": "CD1D34452E4F4B7E",
  "PathId": "E59072E420E51506-CD1D34452E4F4B7E"
},
{
  "Text": "The patient's psychosocial status (eg, depression screening) is assessed and
addressed.",
  "Id": "0EBA06DFCA82FC7A",
  "PathId": "E59072E420E51506-0EBA06DFCA82FC7A"
},
{
  "Text": "The patient's risk for falls is assessed and addressed.",
  "Id": "86D628156332688F",
  "PathId": "E59072E420E51506-86D628156332688F"
},
{
  "Text": "The patient's smoking and tobacco use is assessed and addressed.",
  "Id": "4E9E7D2794123666",
  "PathId": "E59072E420E51506-4E9E7D2794123666"
},
{
  "Text": "The patient's readmission risk is assessed and addressed.",
  "Id": "3F8D32C850EA4893",
  "PathId": "E59072E420E51506-3F8D32C850EA4893"
}
]
},
{
  "Text": "CMS Meaningful Measures include[[Cit:260126]]:",
  ...
}
```

4.15.3.11 Note required

The Indication endpoints in API Version 2 include a new attribute called "NoteRequired" for supplemental Medicare criteria-related indications. This attribute is used to indicate that a note is needed, and data for it must be provided. Without a clinical justification, the indication cannot be saved. The note information should be saved in the integrating solution and not provided in the API call.

The following example shows the Supplemental Medicare Criteria section with as it appears in the ISC guideline content when NoteRequired = *true*.

```
,
  "SectionPath": "/Care_Planning_-_Inpatient_Admission_and_Alternatives/Clinical_Indications_for_
Admission_to_Inpatient_Care",
  "Status": "Not Met",
  "CustomGuideline": false,
  "CustomGuidelineDisclaimer": ""
},
{
  "Heading": "Supplemental Medicare Criteria",
  "Paragraph": "These criteria are to be considered for Medicare patients if none of the standard
```

```
Clinical Indications for Admission to Inpatient Care apply. ",
  "Content": [
    {
      "Status": "Not Met",
      "Text": "Patient with Medicare coverage requires inpatient admission, as indicated by 1 or
more of the following[[Cit:2814737]]:",
      "ListLogic": "1 or more of the following",
      "Id": "CF7468E49DC1BB2F",
      "PathId": "CF7468E49DC1BB2F",
      "CitationLinks": [
        {
          "Id": "2814737",
          "DisplayValue": "1",
          "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/citation/2814737"
        }
      ],
      "ContentItems": [
        {
          "Text": "Admitting clinician expects patient to require hospital care for less than 2
midnights but, based on complex medical factors documented in medical record, judges that inpatient
care is necessary (case-by-case exception). The medical record must contain sufficient documentation
to make clear the rationale for the exception.",
          "Id": "54E3B54F41AC2424",
          "PathId": "CF7468E49DC1BB2F-54E3B54F41AC2424",
          "NoteRequired": true
        },
        {
          "Text": "Patient has need for intubation and mechanical ventilation that is new (ie, did
not present to hospital already on mechanical ventilation).",
          "Id": "34FD5FE87EF6CCBF",
          "PathId": "CF7468E49DC1BB2F-34FD5FE87EF6CCBF"
        },
        {
          "Text": "Treatment plan for hospital admission includes procedure designated by CMS as
inpatient only (ie, on Inpatient Only List).",
          "Id": "AFAFB98C7BD27B5F",
          "PathId": "CF7468E49DC1BB2F-AFAFB98C7BD27B5F"
        },
        {
          "Text": "Patient has already received medically necessary hospital care that meets 2-
midnight benchmark (excluding activities such as triage/intake, delays in provision of care, or time
added due to patient or family convenience). The medical record must contain sufficient documentation
to make clear the medical necessity for hospital care across 2 or more midnights.",
          "Id": "C6F9DA04FF6079BE",
          "PathId": "CF7468E49DC1BB2F-C6F9DA04FF6079BE",
          "NoteRequired": true
        }
      ]
    }
  ],
}
```

4.15.4 Rapid Review Guidelines

The Rapid Review Guidelines (RRG) content is not available with the Indication endpoints in Version 1 of the Informed by MCG API. The GET response is empty as follows.

Version 1: Without RRG content


```
{
  "Heading": "Clinical Indications for Admission",
  "Content": [
    {
      "Text": "Angina",
      "PathId": ""
    }
  ],
  "SectionPath": "/Clinical_Indications_for_Admission",
  "CustomGuideline": false
}
```

In API Version 2, the RRG content appears in the GET response.

From the Clinical Indications for Admission section, you can view the Observation Care and Inpatient Care side-by-side heading structures in the response. Clinicians use this "quick view" to manage patient admissions.

Inpatient & Surgical Care > Rapid Review Guidelines > Adult > Cardiology > Angina RRG (M-40-RRG) - Discharge Readiness - Extended Stay - Discharge Destination - Footnotes - Codes	
Clinical Indications for Admission Return to top of Angina RRG - ISC	
Observation Care - Observation is appropriate for patient with angina and 1 or more of the following^{[A][B][C]}: - Suspected acute coronary syndrome (eg, new symptoms or acute worsening of stable angina symptoms) and 1 or more of the following:	Inpatient Care [Expand All / Collapse All] - Admission is indicated by 1 or more of the following: - Angina necessitating acute intervention (angiography with possible revascularization)

Version 2: With RRG content

```
{
  "Sections": [
    {
      "Heading": "Observation Care",
      "Content": [
        {
          "Text": "Observation is appropriate for patient with angina and 1 or more of the following",
          "ListLogic": "1 or more of the following",
          "Id": "7DE97EC7EB48DC5E",
          "PathId": "7DE97EC7EB48DC5E",
          "FootnoteLinks": [
            {
              "Id": "f0428109v_23_1_75",
              "Link": "https://api-EndpointURL.com/api/v2/GA/27.0/ISC/001rrg_03832/footnote/f0428109v_23_1_75",
              "DisplayValue": "[A]"
            }
          ],
          ...
        }
      ],
      ...
    },
    {
      "Heading": "Inpatient Care",
      "Content": [
        {
          "Text": "Admission is indicated by 1 or more of the following:",
          ...
        }
      ],
      ...
    }
  ],
  ...
}
```

```
"ListLogic": "1 or more of the following",
"Id": "D40F5C2F1BA84A8A",
"PathId": "D40F5C2F1BA84A8A",
"ContentItems": [
  {
```

4.15.4.1 Extended notice information

The Indications endpoint in API Version 2 includes extended notice information for the RRG criteria when present in the response. This information can be displayed in integrated solutions. The RRG response returns the notice under the **Text** element in two different ways, depending on whether there are multiple indication sections or just one.

The first example shows how the response displays the notice information from the S-424-RRG content for one indication section.

```
{
  "Heading": "Clinical Indications for Procedure and Care",
  "Content": [
    { ... },
    {
      "Text": "See Elbow Fracture, Open Treatment Optimal Recovery Guideline for complete guideline and further supporting content such as Supplemental Medicare Criteria (ie, CMS' Two-Midnight Rule exceptions), Alternatives to Admission or Procedure, Optimal Recovery Course, and Evidence Summary.",
      "PathId": "",
      "GuidelineLinks": [.....]
    }
  ],
  "Sections": null,
  "SectionPath": "/Clinical_Indications_for_Procedure_and_Care",
  "ContentVersion": "28.0",
  "CustomGuideline": false,
  "CustomGuidelineDisclaimer": ""
}
```

The second example shows how the response displays the notice information from the M-305-RRG content for multiple indication sections.

```
{
  "Sections": [
    {
      "Heading": "Observation Care",
      "Content": [.....],
      "SectionPath": "/Clinical_Indications_for_Admission/Observation_Care"
    },
    {
      "Heading": "Inpatient Care",
      "Content": [.....],
      "SectionPath": "/Clinical_Indications_for_Admission/Inpatient_Care"
    },
    {
      "Heading": "",
      "Content": [
        {
          "Text": "See Tetraplegia, Acute Optimal Recovery Guideline for complete guideline and further supporting content such as Supplemental Medicare Criteria (ie, CMS' Two-Midnight Rule exceptions), Alternatives to Admission or Procedure, Optimal Recovery Course, and Evidence Summary.",
```

```

    "PathId": "",
    "GuidelineLinks": [.....]
  }
],
"SectionPath": "/Clinical_Indications_for_Admission"
}
],
"SectionPath": "/Clinical_Indications_for_Admission",
"ContentVersion": "28.0",
"CustomGuideline": false,
"CustomGuidelineDisclaimer": ""
}

```

4.16 Information endpoints

GET /api/v2/{contentOwner}/{edition}/{product}/InformationReview/{hsim}
 GET /api/v2/{contentOwner}/{edition}/{product}/EvaluationTreatment/{hsim}

There are two Information endpoints in API Version 2: GET InformationReview and GET EvaluationTreatment. Use these endpoints to retrieve information including the clinical indication logic needed for hospitalization reviews and retrieve the Evaluation and Treatment section of the General Recovery Care Guideline (GRG) content including the clinical indication logic.

The following example shows the Evaluation and Treatment section as it appears in the GRG content.

General Recovery Care > Long-Term Acute Care Hospital (LTACH) General Recovery Guidelines > Long-Term Acute Care Hospital (LTACH) Level of Care Guideline (GRG-050)

Long-Term Acute Care Hospital (LTACH) Level of Care Guideline

GRG: GRG-050 (LTACH GRG)

Note: An appropriate Optimal Recovery Guideline (ORG) should be identified and used whenever possible. This General Recovery Guideline (GRG) is intended to aid only in situations in which no ORG appears applicable.

- Care Planning - LTACH Admission and Alternatives
 - Clinical Indications for Admission to LTACH
 - Alternatives to Admission
- Hospitalization
 - General Recovery Course
 - Evaluation and Treatment**
 - Discharge Criteria

The following example shows how the response for the EvaluationTreatment endpoint typically displays.

```

{
  "Heading": "Evaluation and Treatment",
  "Content": [
    {
      "Text": "Common treatments and tests include<a class='citation' data-id='102921'>5</a><a class='citation' data-id='2538612'>31</a><a class='citation' data-id='2522586'>47</a>:",
      "Id": "1FBDFAF17329467",
      "PathId": "1FBDFAF17329467",
    }
  ]
}

```

```
"CitationLinks": [
  {
    ...
  }
]
```

4.17 Length of Stay endpoint

GET /api/v2/{contentOwner}/{edition}/{product}/LengthOfStay/{hsim}

In API Version 2, the LengthOfStay endpoint retrieves the Goal Length of Stay section of a specific guideline. The following example shows the Goal Length of Stay section as it appears in the ISC guideline content.

Inpatient & Surgical Care > Optimal Recovery Guidelines > Cardiology > Heart Failure (M-190)

Heart Failure

ORG: M-190 (ISC)

[Link to Codes](#)

[Print View](#)

- Care Planning - Inpatient Admission and Alternatives
 - Clinical Indications for Admission to Inpatient Care
 - Alternatives to Admission
 - Alternative Care Planning
- Hospitalization
 - Optimal Recovery Course
 - Goal Length of Stay - 2 days**
 - Extended Stay
 - Hospital Care Planning

The following example shows how the response for the LengthOfStay endpoint typically displays.

```
{
  "Heading": "Goal Length of Stay",
  "Type": "DS",
  "Min": "2",
  "Max": "2",
  "Display": "GLOS",
  "Paragraph": "2 days",
  "SectionPath": "/Hospitalization/Optimal_Recovery_Course",
  "Notices": [
    {
      "Id": "glosdefaultnote",
      "Message": "Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs."
    }
  ]
}
```

4.18 Levels of Care endpoints

GET /api/v2/{contentOwner}/{edition}/{product}/LevelsOfCare/{hsim}

POST /api/v2/{contentOwner}/{edition}/{product}/LevelsOfCare/{hsim}

GET /api/v2/{contentOwner}/{edition}/{product}/AlternativeLevelsOfCare/{hsim}

With the Levels of Care endpoints In API Version 2, you can perform the following tasks:

- Retrieve the Levels of Care Criteria and Intensity of Care Criteria sections including the clinical indication logic. The POST endpoint includes a Status element for each indication that shows whether the patient conditions have been met. MCG evaluates and sets the value of the status.
- Retrieve the Alternative Levels of Care section of a specific guideline.

The following example shows the Intensity Of Care Criteria as it appears in the Inpatient & Surgical Care > Rapid Review Guidelines.

Inpatient & Surgical Care > Rapid Review Guidelines > Neonatal > Neonatology > Neonatal Jaundice RRG (P-265-RRG)

Neonatal Jaundice RRG

RRG: P-265-RRG (ISC)

[Link to Codes](#)

- [Clinical Indications for Admission](#)
- [Intensity of Care Criteria](#)
- [Hospitalization](#)

From the Intensity of Care Criteria chart, each care-level section (4-1) is evaluated independently of each other to determine whether the patient conditions have been met.

Inpatient & Surgical Care > Rapid Review Guidelines > Neonatal > Neonatology > Neonatal Jaundice RRG (P-265-RRG)

Intensity of Care Criteria			
Return to top of Neonatal Jaundice RRG - ISC			
For patient conditions requiring more than routine inpatient care, documentation of need may be done using the Intensity of Care chart.			
Intensive Care - 4	Intermediate Care - 3	Continuing Care - 2	Routine Care - 1
Intensive Care: Intensity of Care Criteria 4 [I-4] provided, as indicated by 1 or more of the following:	Intermediate Care: Intensity of Care Criteria 3 [I-3] provided, as indicated by 1 or more of the following:	Continuing Care: Intensity of Care Criteria 2 [I-2] provided, as indicated by 1 or more of the following:	Routine Care: Intensity of Care Criteria 1 [I-1] provided, as indicated by 1 or more of the following:

The following example shows how the response for the GET LevelsOfCare endpoint displays the side-by-side heading structures.

```
{
  "Sections": [
    {
      "Heading": "Intensive Care - 4",
      "Content": [
        {
          "Text": "Intensive Care: Intensity of Care Criteria 4[[Foot:fdef_05356_0001]][[Foot:fdef_05356_0002]][[Foot:fdef_05356_0003]] provided, as indicated by 1 or more of the following:",
          "ListLogic": "1 or more of the following",
          "Id": "09B0199C859C20C6",
          "PathId": "09B0199C859C20C6",
          "Definitions": [
            {
              ...
            }
          ]
        },
        {
          "Heading": "Intermediate Care - 3",
          ...
        }
      ]
    }
  ]
}
```

```

"Content": [
  {
    "Text": "Intermediate Care: Intensity of Care Criteria 3[[Foot:fdef_05356_0002]][[Foot:fdef_05356_0003]][[Foot:fdef_05355_0001]] provided, as indicated by 1 or more of the following:",
    "ListLogic": "1 or more of the following",
    "Id": "D995A63C160013B9",
    "PathId": "D995A63C160013B9",
    "Definitions": [
      {
        ...
      }
    ],
  },
  {
    "Heading": "Continuing Care - 2",
    "Content": [
      {
        "Text": "Continuing Care: Intensity of Care Criteria 2[[Foot:fdef_05356_0002]][[Foot:fdef_05356_0003]][[Foot:fdef_05354_0001]] provided, as indicated by 1 or more of the following:",
        "ListLogic": "1 or more of the following",
        "Id": "32632A4D1DBF0BA6",
        "PathId": "32632A4D1DBF0BA6",
        "Definitions": [
          {
            ...
          }
        ],
      },
      {
        "Heading": "Routine Care - 1",
        "Content": [
          {
            "Text": "Routine Care: Intensity of Care Criteria 1[[Foot:fdef_05356_0002]][[Foot:fdef_05356_0003]][[Foot:fdef_05353_0001]] provided, as indicated by 1 or more of the following:",
            "ListLogic": "1 or more of the following",
            "Id": "9A54EF1E99CF6EAB",
            "PathId": "9A54EF1E99CF6EAB",
            "Definitions": [
              {
                ...
              }
            ],
          }
        ],
      }
    ],
  }
]

```

4.19 Patient Education endpoints

GET /api/{contentOwner}/{edition}/{product}/PatientEducation/{hsim}
POST /api/{contentOwner}/{edition}/{product}/PatientEducation

You can retrieve the patient education materials, if available, for a specific guideline or product. This endpoint retrieves only the latest edition. Minor versions are available through other endpoints.

The patient education materials are encoded in PDF format and must be decoded before use. The HSIM code can be retrieved from your search results. For more information, see the [Search endpoints](#).

A single patient education document is returned. The following image shows a sample of a patient education PDF.

Preoperative Information

Repair of Abdominal Aortic Aneurysm

You are having surgery to repair a weak spot of the aorta in your stomach area (abdomen). The aorta is the large artery that carries blood from the heart. The surgeon will replace the weak spot with a small tube called a graft.

This handout tells you what you need to do to get ready for your surgery. Ask your doctors or nurses if there are things you do not understand.

Many people can leave the hospital within 5 days after surgery. Other people need more time in the hospital. Recovery time varies from person to person.

 **During the weeks before surgery**

- You will need to see your regular doctor for a checkup before surgery.
- You may need to have tests done before the surgery, such as:
 - EKG (electrocardiogram)
 - Blood tests
 - Chest x-ray
 - Other tests your doctor thinks you need
- You may be given information on changes to your diet or lifestyle you may need to make after surgery.
- You should not drink alcohol in the weeks leading up to surgery. Your doctor will tell you when you should stop.

4.20 Ping endpoint

`GET /api/ping`

The Ping endpoint is used to test network connectivity. If successful, a PONG response is returned.

4.21 Prevention endpoint

`GET /api/v2/{contentOwner}/{edition}/{product}/Prevention/{hsim}`

In API Version 2, the Prevention endpoint retrieves the Prevention section of a specific guideline. The following example shows the Prevention section as it appears in the ISC guideline content.

Inpatient & Surgical Care > Common Complications and Conditions > Alcohol and Psychoactive Substance Withdrawal: Common Complications and Conditions (CCC-035)

Alcohol and Psychoactive Substance Withdrawal: Common Complications and Conditions

CCC: CCC-035 (ISC)
[Link to Codes](#)

- **Prevention**
- Clinical Indications for Inpatient Care
- Alternatives to Inpatient Care

The following example shows how the response for the Prevention endpoint typically displays.

```
{
  "Heading": "Prevention",
  "Content": [
    {
      "Text": "Prevention issues and possible preventive measures include:",
      "Id": "2D25FD1EFB497922",
      "PathId": "2D25FD1EFB497922",
      "ContentItems": [
        {
          "Text": "At admission, identify patient at risk[[Foot:f9900037t_1316_1_28]][[Foot:f9900037t_28_1_28]] for substance withdrawal by history and symptoms; examples include[[Cit:258612]][[Cit:164241]][[Cit:228084]][[Cit:2515856]]:",
          "ListLogic": "Examples Include",
          "Id": "0195FC25E16D1E93",
          "PathId": "2D25FD1EFB497922-0195FC25E16D1E93",
          "CitationLinks": [
            {
              "Id": "258612",
              "DisplayValue": "1",
              "Link": "https://api-EndpointURL.com/api/v2/GA/27.0/citation/258612"
            }
          ],
          ...
        }
      ]
    }
  ]
}
```

4.22 Product endpoint

GET /api/product

The Product endpoint retrieves a list of the content bundles that are available for the client ID and client key provided by MCG. Use this endpoint to see what products are licensed on the client. You can also check a license.

This endpoint returns bundle names that have a 1:1 association with MCG content products.

4.23 Recovery Course endpoints

There are three endpoints: GET RecoveryCourse endpoint, GET RecoveryStage endpoint, and POST RecoveryStage endpoint. With these endpoints, you can retrieve the content of a Recovery Course section of a specific guideline. The supported section titles are Recovery Course, General Recovery Course, General Treatment Course, and Optimal Recovery Course.

4.23.1 GET RecoveryCourse endpoint

GET /api/{contentOwner}/{edition}/{product}/recoverycourse/{hsim}

This endpoint returns the entire content of a Recovery Course section of a specific guideline. The information is view-only; there is no corresponding POST endpoint to calculate patient progress within the Recovery Course.

4.23.2 RecoveryStage endpoints

GET /api/{contentOwner}/{edition}/{product}/recoverycourse/{hsim}/recoverystage/{recoveryStage}
POST /api/{contentOwner}/{edition}/{product}/recoverycourse/{hsim}/recoverystage/{recoveryStage}

They are used in conjunction with the General Treatment Course guideline section of a Recovery Facility Care, Home Care, or Behavioral Health Care guideline. The General Treatment Course describes the level of care, interventions, and anticipated patient progress for each stage of recovery facility or outpatient recovery under ideal circumstances. A stage within a treatment course can span many days.

- The GET RecoveryStage endpoint retrieves the content of the General Treatment Course. You have the option to retrieve a single stage or the entire content of a General Treatment Course.
- The POST RecoveryStage endpoint evaluates whether the patient has met the requirements to move to the next stage in the course. To retrieve the entire treatment course, leave the recoveryStage parameter empty.

The RecoveryStage parameter is required for each POST message. The POST message processes only the information for a single stage within a General Treatment Course. If you want to evaluate more than one stage, you must submit a separate POST message for each stage.

The typical workflow when using the RecoveryStage endpoints is:

1. Retrieve the content of the General Treatment Course using the GET endpoint.
 - a. Retrieve all stages using the RecoveryCourse endpoint.
 - b. Retrieve a single stage using the RecoveryStage endpoint. To do this, specify the numeric value of the stage (1, 2, or 3) in the recoveryStage parameter.
2. Display the retrieved General Treatment Course content to a user as a selectable list of care elements and milestones.
3. Submit the selected care elements and milestones to MCG using the POST endpoint.
4. MCG returns a status indicating whether the patient is ready to move to the next stage in the General Treatment Course.

The entries in a General Treatment Course can contain footnotes, annotations, citations, and definitions.

4.23.3 About the Optimal Recovery Course

Some of the care guidelines include a section that describes day-by-day interventions and recovery milestones that would occur under ideal circumstances. The content in these sections is occasionally also referred to as a progression.

Inpatient care is described using a Recovery Course. Details for each day of expected patient care (the goal length of stay) is provided in the Recovery Course.

The following image shows an example of an Optimal Recovery Course.

Optimal Recovery Course						
Return to top of Heart Failure - ISC						
Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	- ICU^[1] or intermediate care^[2] after emergency treatment - Readmission Risk Assessment - Extended Stay Risk Assessment - Discharge planning	- Clinical Indications met^[3] - Tachypnea, edema, and dyspnea - Possible hypoxia and hypotension	Initial bed rest	- Parenteral medications - Oral hydration - Low-salt diet	- ECG, CXR, ABG - Cardiac biomarkers, BNP - Possible pulmonary artery catheter^[4] - Oxygen - Echocardiogram - Digoxin level if previously prescribed - Electrolytes - Weight - Possible fluid restriction	- IV diuretics - ACE inhibitor or ARB - Beta-blocker^[5] - Possible aldosterone antagonist - Possible IV vasodilator (eg, nitrates, hydralazine)
2	Intermediate care or floor	- Hemodynamic stability - Mental status at baseline - MI excluded - Cardiac rate and rhythm acceptable - Oxygenation at baseline or improved - Pulmonary edema absent or	Advance activity as tolerated	- Oral or parenteral medications - Low-salt diet - Adjust fluid restriction	- Pulmonary catheter absent - Weight - Possible electrolytes^[6] - Possible CXR, ECG, BNP - Oxygen	- Taper parenteral medication and optimize oral medication - Diuretics - ACE inhibitor or ARB - Beta-blocker - Possible aldosterone antagonist - Possible nitrates, digoxin, hydralazine

When working on an Optimal Recovery Course, the patient's progress is tracked day by day. When all recovery milestones for a care day have been met, the patient is considered ready to move on to the next care day. When the patient has completed all of the care days for the guideline, they are ready for discharge.

The entries in an Optimal Recovery Course can contain footnotes, annotations, citations, and definitions.

4.23.3.1 Care day versus care date

The terms, *care day* and *care date*, are used with Optimum Recovery Courses. They have different meanings.

A care day, also called a guideline care day, refers to a day as specified in the Optimum Recovery Course. For example, an Optimum Recovery Course might describe three days; each one of those days is considered to be a care day.

The care date is the calendar date on which patient care was given or recovery milestones met. The care date does not always correspond to the care day. If a patient recovers more slowly than expected, it is possible that multiple care dates will be recorded against a single guideline care day. Likewise, if a patient recovers more quickly than expected, a single care date can comprise multiple guideline care days.

4.23.4 About the General Treatment Course

Outpatient and long-term care are described using a General Treatment Course. A General Treatment Course is very similar to an Optimal Recovery Course, except that it tracks progress based on a broader criteria of stages. Each stage in a General Treatment Course can span multiple days.

The following image shows how the General Treatment Course appears in the M-5190 guideline, Recovery Facility Care Heart Failure.

Stage	Clinical Status	Interventions
1	- Clinical indications for admission met - Recovery Facility Clinical Assessment ↗ RFC complete QM - Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment ↗ SR complete - Psychosocial Assessment ↗ SR complete - Medication reconciliation complete. See Medication Reconciliation Tool ↗ SR for more information. QM - Hospital readmission risk factor evaluation(7)(8)(9) - Skilled needs identified	- Transition of care planning initiated with evaluation for next level of care(10)(11) - Interdisciplinary care plan established and implemented - Chronic disease condition management as indicated(12)(13) - Fall risk management QM - IV infusion (eg, fluids, antibiotics, parenteral nutrition) - Medication management QM (14) - Nutrition management(15)(16) - Patient and caregiver education(17)(18) - Pressure ulcer risk management QM - Psychosocial issues identified and management initiated, as needed - Respiratory care(19)(20) - Physical therapy (PT) referral and evaluation - Occupational therapy (OT) referral and evaluation
2	- Therapeutic response to interventions - Rehabilitation program established with patient participation, or patient is unwilling or unable to participate. - Skilled services cannot be managed at lower level of care. - Physical therapy (PT) progression QM - Occupational therapy (OT) progression QM	- Care plan continues. - Ongoing assessment of clinical needs - Transition of care planning continues with evaluation for next level of care. (21) - Continue patient and caregiver education.
3	- Medication regimen established and reconciliation completed - Medical status stable for patient's condition and manageable at lower level of care - Immunizations implanted, continued, or functioning normally and manageable at lower level of care	- Transition of care planning completed - Discharge from recovery facility care

Patient data may be recorded based on a care date, or the date on which the care was delivered, or patient recovery was assessed. Each stage can contain as many care dates as needed for the patient to complete the recovery milestones for the stage. When all stages are complete, the patient is ready for discharge.

Informed by MCG for Utilization Management web service provides two endpoints for use with recovery stages:

- The GET RecoveryStage endpoint which retrieves the content of a General Treatment Course. You can retrieve the entire recovery course or just one stage of the course as wanted.
- The POST RecoveryStage endpoint evaluates completed patient recovery milestones and determines whether the patient is ready to proceed to the next care stage.

4.24 References endpoint

```
GET /api/v2/{contentOwner}/{edition}/{product}/References/{hsim}
```

In API Version 2, the References endpoint retrieves the References section of a specific guideline. The following example shows the References section as it appears in the ISC guideline content.

Inpatient & Surgical Care > Optimal Recovery Guidelines > Cardiology > Heart Failure (M-190)

- Discharge Destination
- Evidence Summary
 - Alternatives
 - Hospitalization
 - Length of Stay
- **References**
- Footnotes
- Codes

The following example shows how the response for the References endpoint typically displays.

```
{
  "Citations": [
```

```
{
  "Id": "116500",
  "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/citation/116500",
  "CitationText": "Yancy CW, et al. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. Circulation 2013;128(16):e240-e319. DOI: 10.1161/CIR.0b013e31829e8776. (Reaffirmed 2020 Apr) ",
  "AbstractText": "https://www.URL/pubmed/23741058"
},
{
  "Id": "199014",
  "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/citation/199014",
  "CitationText": "Yancy CW, et al. 2017 ACC/AHA/HFSA focused update of the 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Failure Society of America. Circulation 2017;136(6):e137-e161. DOI: 10.1161/CIR.0000000000000509. (Reaffirmed 2020 Jan) ",
  "AbstractText": "https://www.URL/pubmed/28455343"
},
{
  "Id": "259522",
  "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/citation/259522",
  "CitationText": "Hollenberg SM, et al. 2019 ACC expert consensus decision pathway on risk assessment, management, and clinical trajectory of patients hospitalized with heart failure: a report of the American College of Cardiology Solution Set Oversight Committee. Journal of the American College of Cardiology 2019;74(15):1966-2011. DOI: 10.1016/j.jacc.2019.08.001. (Reaffirmed 2020 Apr) ",
  "AbstractText": "https://www.URL/pubmed/31526538"
},
{
  "Id": "203553",
  "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/citation/203553",
  "CitationText": "O'Brien JF, Hunter CL. Heart failure. In: Walls RM, et al., editors. Rosen's Emergency Medicine. 9th ed. Philadelphia, PA: Elsevier; 2018:971-986.",
  "AbstractText": ""
},
{
  "Id": "232674",
  "Link": "https://api-EndpointURL.com/api/v2/GA/28.0/citation/232674",
  "CitationText": "Masarone D, et al. Pediatric heart failure: a practical guide to diagnosis and management. Pediatrics and Neonatology 2017;58(4):303-312. DOI: 10.1016/j.pedneo.2017.01.001. ",
  "AbstractText": "https://www.URL/pubmed/28279666"
},
}
```

4.25 Reference Links endpoint

GET/api/v2/{contentOwner}/{edition}/{product}/{hsim}/referencelinks

In API Version 2, the `referencelinks` endpoint retrieves the reference links based on the specified guideline/HSIM value. The response for this endpoint is like the Search response but limited to a specific HSIM. Use this endpoint in conjunction with the [Table of Contents endpoint](#) to create a navigable TOC.

The following example shows the reference links for Alternatives to Procedure and Evidence Summary as they appear in the AC guideline content.

Content Guide to Ambulatory Care

- Ambulatory Care Description
- Ambulatory Care Title and Code
- Ambulatory Care Guidelines
 - Clinical Indications
 - "Current Role Remains Uncertain"
 - **Alternatives to Procedure**
 - **Evidence Summary**
 - Evidence Summary - Background
 - Evidence Summary - Criteria
 - Evidence Summary - Inconclusive or Non-Supportive Evidence

The following example shows how the response typically displays.

```
[
  {
    "ContentTitle": "Alternatives to Procedure",
    "StructuredContentType": "Alternatives",
    "ReferenceLink": "https://api-mcg.websites.net/api/v2/GA/27.0/AC/Alternatives/006ac11_075"
  },
  {
    "ContentTitle": "Evidence Summary",
    "StructuredContentType": "EvidenceSummary",
    "ReferenceLink": "https://api-mcg.websites.net/api/v2/GA/27.0/AC/EvidenceSummary/006ac11_075"
  },
  ...
]
```

4.26 Regions endpoint

GET /api/v2/{contentOwner}/{edition}/regions

In API Version 2, the Regions endpoint retrieves the list of possible LCD regions to filter MCR search results. The following example shows how the response for the Regions endpoint typically displays.

```
[
  "Alabama",
  "Alaska",
  "American Samoa",
  "Arizona",
  "Arkansas",
  "California - Entire State",
  "California - Northern",
  "California - Southern",
  "Colorado",
  "Connecticut",
  "Delaware",
  "District of Columbia",
  "Florida",
  ...
]
```

4.27 Related Codes Search endpoint

GET /api/{contentOwner}/{edition}/{product}/RelatedCodes/{hsim}

The GET RelatedCodesSearch endpoint identifies the codes (ICD, ICP, DSM, or CPT/HCPSCS) that are associated with a particular guideline.

MCG includes the related codes in the guideline content. The following example shows how the related codes are displayed in the M-190, Heart Failure, guideline.

Codes
ICD-10 Diagnosis: I09.81, I11.0, I13.0, I13.2, I50.1, I50.20, I50.21, I50.22, I50.23, I50.30, I50.31, I50.32, I50.33, I50.40, I50.41, I50.42, I50.43, I50.9, T86.22 [Hide]
ICD-9 Diagnosis: 398.91, 402.01, 402.11, 402.91, 404.01, 404.03, 404.11, 404.13, 404.91, 404.93, 428.0, 428.1, 428.20, 428.21, 428.22, 428.23, 428.30, 428.31, 428.32, 428.33, 428.40, 428.41, 428.42, 428.43, 428.9, 998.01 [Hide]

4.28 Search endpoints

With the Search endpoints, you can retrieve a list of guidelines that may be relevant to a patient case. Each search result includes an HSIM code, or identifier, which allows you to retrieve the content of the guideline.



With API version 2 of the Search endpoints, you can include a minor edition (e.g., "27.1") in the search and receive that minor edition (including the latest version of guidelines if they were not updated in that edition).

4.28.1 Search endpoint (single guideline product)

`GET /api/{contentOwner}/{edition}/search`

This endpoint is the most used guideline search and provides the same functionality as the Search page in MCG solutions, limited to a single bundle. Using this endpoint, you can search by keyword, guideline code, procedure, or diagnostic code. Multiple search values can be supplied as well as AND/OR logic. The optional parameter **guidelineType** filters the results by the requested guideline type separated by commas or spaces, for example, ACG, NCG, ORG, ORG, ...

To filter the search results, use optional filters: **searchDate** and **guidelineType**.

4.28.2 Search endpoint (multiple guideline products)

`GET /api/{contentOwner}/{edition}/{product}/search`

This endpoint provides the same functionality as the Search page in MCG solutions, and searches across multiple MCG content products. You can retrieve the supported list of bundles using the Product endpoint. Using this endpoint, you can search by keyword, guideline code, or procedure or diagnostic code. Multiple search values can be supplied as well as AND/OR logic.

To filter the search results, use the optional filters: **searchDate** and **guidelineType**.

4.28.3 Search endpoint (single AutoAuth guideline product)

`GET /api/{contentOwner}/{edition}/{product}/AuthorizationSearch`

This endpoint is limited to a single bundle based on your AutoAuth guideline rules. You can search by keyword, guideline code, or procedure or diagnostic code. Multiple search values can be supplied as well as AND/OR logic.

4.28.4 Search endpoint (multiple AutoAuth guideline products)

GET /api/{contentOwner}/{edition}/AuthorizationSearch

This endpoint searches across multiple MCG content products based on your AutoAuth guideline rules. You can search by keyword, guideline code, or procedure or diagnostic code. Multiple search values can be supplied as well as AND/OR logic.



To activate and use the AuthorizationSearch endpoints in your preauthorization workflows, contact your account manager or sales representative for assistance.

4.29 Table of Contents endpoint

GET /api/V2/{contentOwner}/{edition}/TableOfContents

The TableOfContents endpoint in API Version 2 lets you retrieve the Table of Contents (TOC) from a specific MCG licensed solution, such as Chronic Care. The MCG solution is limited to a single bundle and content edition/version.

The following example shows part of the TOC for CCG. The titles are organized into top-level sections and subsections as a list. You can expand the pre-populated sections (such as Start of Care) and view a guideline, cross-reference, or an external website by selecting the appropriate hyperlink.

Chronic Care

- Introduction
 - [Content Guide to Chronic Care](#)
- Methodology Information
 - [Summary of Guideline Development Policies and Procedures](#)
- High Intensity Disease Management
 - Enrollment
 - [Enrollment](#)
 - Social Determinants of Health
 - [Caregiver Resources Evaluation](#)
 - [Community Resources](#)
 - [Education Management](#)
 - [Financial Status and Benefits](#)
 - [Health Literacy](#)
 - [Safety](#)
 - [Screening for Health-Related Social Needs](#)
 - [Social Support](#)
 - [Transportation](#)
- [Start of Care](#)
 - [Self-Management](#)

The following example shows how the response displays the TOC for CCG. The titles and parts of the TOC retrieved by the TableOfContents endpoint are defined by the **Items** instance as shown.

```
{
  "Title": "Chronic Care",
  "Product": "CCG",
  "VersionMajor": 26,
  "VersionMinor": 0,
  "Items": [
    {
      "Name": "Introduction",
      "Expanded": true,
      "LinkType": "None",
      "Items": [
        {
          "Name": "Content Guide to Chronic Care",
          "FileId": "ccmintro",
          "Expanded": false,
          "Hsim": "008ccmintro",
          "LinkType": "Content"
        }
      ]
    },
    ...
  ],
  "Items": [
    {
      "Name": "Start of Care",
      "SectionSeparator": true,
      "Expanded": false,
      "Collapsible": true,
      "LinkType": "None",
      "Items": [
        {
          "Name": "Start of Care Assessment, Adult",
          "FileId": "iccg_207",
          "Expanded": false,
          "Hsim": "008iccg_207",
          "LinkType": "Content"
        }
      ],
      ...
    }
  ]
}
```

4.29.1 Rendering the TOC

The TOC document contains all top-level metadata for the JSON document retrieved by the TableOfContents endpoint (InformedBy.Api.Models.TableOfContents), as well as a collection of Item instances. The Item model is a recursive structure, where each individual instance represents a header or link, when combined with all other Items, make up the structure of the TOC.

4.29.1.1 Document model

The following properties table describes the accepted values.

Property name	Data type	Is required?	Value description
Footer	String array	Yes	A collection of strings, each one representing a single line of the MCG footer.
Items	Item array	Yes	A collection of Item instances representing the structure of the TOC document on the page.
Notices	String array	No	A collection of custom notes that you can display at the top of the TOC page when viewing the document as a whole.
Product	String	Yes	The product code of the represented TOC document (ex: CCG).
Title	String	Yes	The title of the represented TOC document (ex: Chronic Care).
VersionMajor	Numeric	Yes	The product solution's major version (ex: 26)
VersionMinor	Numeric	Yes	The product solution's minor version (ex: 1)

4.29.1.2 Item model

The following properties table describes the accepted values.

Property name	Data type	Is required?	Value description
Collapsible	Boolean	No	Determines whether an item is rendered with an expandable (+) or collapsible (-) button beside it. Default: False
ColumnSeparator	Boolean	No	Determines whether an item is rendered at the top of a new column. Default: False
CrossProduct	String	No	When included, refers to the product code of the document that the item links to.
Expanded	Boolean	Yes	Specifies whether the children within an item instance are visible by default. Default: False
FileId	String	No	An identifier (file name or unique ID) that changes based on the context of the item provided by the LinkType property as follows: - Content: A relative path representing the location of the content document (unused during rendering). - Navigation: Identifier (unused during rendering). - External: Any relative or absolute URL pointing to a dynamic page (ex: Benchmarks and Data links). - Pdf: The PDF filename. - Static: Any URL that will be rendered directly to the page with no extra processing.
HSIM	String	No	When included, represents the unique identifier for a guideline. Accepted value: HSIM code

Property name	Data type	Is required?	Value description
IsCustom	Boolean	No	Determines whether or not the item displays custom content for guidelines or medical policies created using GMM or the Table of Contents Editor tool. Default: False
IsDeactivated	Boolean	No	Determines whether the item is deactivated in the Guideline Modification Module or the Table of Contents Editor tool. The entire sub-tree represented by the item should be excluded from rendering. Default: False
Items	Item array	No	The child item nodes within an item.
LinkType	String	Yes	Determines the context of the item and changes the way the properties are used for rendering items within the TOC structure. Accepted values: None, Content, Navigation, External, Pdf, or Static
Name	String	Yes	Text name displayed for an item.
Notices	String array	No	A collection of custom notes that you can display at the top of the document when the item is the current top-level item.
SectionSeparator	Boolean	No	Determines whether the current item is immediately preceded by an empty line. Default: False

4.29.2 Item rendering

The following table describes how the item rendering process works.

Item property	Property value	Actions
IsDeactivated	True	Complete sub-tree processing.
IsDeactivated	False	Evaluate ColumnSeparator .
ColumnSeparator	True	Begin a new column and evaluate SectionSeparator .
ColumnSeparator	False	Evaluate SectionSeparator .
SectionSeparator	True	Insert extra line-break before item and evaluate LinkType .
SectionSeparator	False	Evaluate LinkType .
LinkType	Content	Link to guideline using HSim property and evaluate if current item has child items.
LinkType	Static	Construct link using URL in FieldId property and evaluate if current item has child items.
LinkType	External	Construct link using URL in FieldId property and evaluate if current item has child items.

Item property	Property value	Actions
LinkType	PDF	Construct link using ODF and filename in FieldId property and evaluate if current item has child items.
LinkType	None	Render text header using Name field and evaluate if current item has child items.
LinkType	Navigation	Evaluate CrossProduct .
CrossProduct	Yes	Evaluate if FieldId has a value.
CrossProduct	No	Link to page where this item is top-level and evaluate if current item has child items.
FieldId	string value present	Link to matching node of product document and evaluate if current item has child items.
FieldId	string value not present	Link to matching product document and evaluate if current item has child items.
Has child items?	Yes	Evaluate Collapsible .
Has child items?	No	Complete subtree processing.
Collapsible	True	Render expand/collapse buttons and evaluate Expanded .
Collapsible	False	Evaluate Expanded .
Expanded	True	Enumerate the child Items collection.
Expanded	False	Complete sub-tree processing.

4.30 Version endpoint

```
GET /api/{ContentOwner}/version
GET /api/v2/{ContentOwner}/version
```

The Version endpoint identifies the latest edition/version of the MCG guideline content that is available to the content owner. It also specifies the software version number of the MCG web service currently in use.

The version number syntax is *edition.version*. The initial release of a content product is always version 0, such as 27.0. If a content product is revised and re-released, the version number is incremented (27.1). It is possible to concurrently license more than one edition/version of guideline content.

The Version endpoint in Version 2 of the API also identifies the minor edition/version of the guideline content that is available. The version number syntax is *major.minor*. A response of 27.2 indicates that you are working in the second minor edition/version of the 27th Edition guideline content. The following example shows how the response typically displays.

```
{
  "LatestEdition": "27.2",
  "ApplicationVersion": "1.0.0.0"
}
```

4.31 Search tips and examples

In CareWeb, you can search for guideline data in different ways.

4.31.1 Tips for searching by code

When searching by ICD-10 diagnosis, ICD-10 procedure, DSM-5 diagnosis, and CPT®/HCPCS procedure codes, you can:

- Include any leading zeros in a code, such as 0016070.
- Enter diagnosis-related group (DRG) codes in the following format: drg 100.
- Use a partial code or a wildcard to represent one or more characters in a code and specify multiple search terms.

Examples:

Keyword search option	(example)	returns guidelines containing
First two or more characters of a code (must be at least two characters)	I5	codes I50.1 through I52
Last two or more characters of a code (use an asterisk in place of the preceding characters in the code)	*0.1	codes ending in "0.1": I50.1, A40.1, G00.1, and so on
A question mark in place of a single character	I50.?1	codes I50.21, I50.31, I50.41, and so on
Two codes joined with "or"	I5* or I1*	codes beginning with either "I5" or "I1", such as I50.1, I51.7, I13.2, and so on
Codes grouped in parentheses (conducts a search with multiple criteria)	(I5* (*.30 or *.40))	codes beginning with "I5" and a suffix of either ".30" or ".40," such as I50.30 and I50.40

4.31.2 Tips for searching by word

You can use a partial word, use a wildcard to represent one or more characters in a word, and specify multiple search terms.

Examples:

Keyword search option	(example)	returns guidelines containing
First two or more letters of a word (must be at least two characters)	ca	a word starting with "ca" (such as care, cardiac, cannula, cause, or caution)
Last two or more letters of a word (use an asterisk in place of the preceding letters)	*nia	a word ending in "nia" (such as pneumonia, schizophrenia, or hernia)
A question mark in place of a single character	ca?e	the words "case," "car," "careful," and so on

Keyword search option	(example)	returns guidelines containing
A word enclosed in quotation marks	"care"	the four-letter word "care." The results exclude the word "careful."
More than one word enclosed in quotation marks	"heart failure"	the complete phrase "heart failure" exactly as entered
Two words joined with "or"	heart or failure	either the word "heart" or "failure"
Words or phrases grouped in parentheses. (conducts a search for multiple criteria)	("failure" (renal or heart))	the word "failure" and either the word "renal" or "heart"

4.31.2.1 Considerations

Keep in mind:

- If you use a partial word, provide as many characters to match as you can; too few characters may result in a wider range of results than you want.
- If you are not sure how to spell a word, enter a best guess. If the search feature does not recognize the word, it suggests a possible replacement.
- The search function does not recognize upper or lower case. Typing "Cardiac," "cardiac," or "cARDIac" all return the same results.
- The search function does not recognize punctuation marks. Typing "cardiac," "(cardiac)," "cardiac!," or "cardiac:" return the same results.
- If you search for more than one word, the search looks for all care guidelines containing both words, whether they appear separately, together, or in any order. For example, entering "cardiac electrode" returns all care guidelines that contain both the words "cardiac" and "electrode" in any order.
- Use synonyms, such as "brain," "head," "neuro," or "cerebral," if you do not find the guideline you want in your first search.
- After the care guideline appears, use your browser's Find function to look for specific words or phrases.
- You can search for common clinical abbreviations, such as "CHF" and "TURP."

4.32 Endpoint quick reference

If you want to:	Do this
(Optional) Know the latest edition/version of the care guideline content.	Call the Version endpoint : <code>https://api-EndpointURL.com/api/ga/version</code>
See all available product bundles for my application and select a product to search. Check that the MCG content license information on the client is valid.	Call the Product endpoint : <code>https://api-EndpointURL.com/api/product</code>

If you want to:	Do this
See all available guidelines in the 28th Edition product bundle.	Call the Guidelines endpoint: <pre>https://api-EndpointURL.com/api/ga/28/product/guidelines</pre>
Search for a guideline based on a keyword in a 28th Edition product bundle.	Call the keyword Search endpoint (the example searches for the "diabetes" keyword): <pre>https://api-EndpointURL.com/api/ga/28/product/search search?searchType=keyword&value=diabetes</pre>
Search for a guideline based on a diagnosis or procedure code in a 28th Edition across multiple products.	Call the code Search endpoint (the example searches for the ICD-10 "xxxx" code): <pre>https://api-EndpointURL.com/api/ga/28/search search?searchType=ICD10-D&value=xxxx</pre> The available code types are ICD9-D, ICD10-D, ICD9-P, ICD10-P, DSM-IV, DSM-5, CPT, and HCPCS.

5 Error messages

The following list describes error messages that commonly appear in the endpoint responses.

Message text	Event	Possible causes
An error occurred, please try again or contact the administrator. There is an error in XML document (1, 45).	There was a syntax error in the request XML; the XML was not well formed.	A syntax error in the submitted XML.
Annotation ID must be a number.	The requested Annotation ID contained characters other than numbers.	The value provided for the Annotation ID contains a value other than the digits 0-9.
Annotation was not found.	The requested Annotation ID was not found for the specified guideline.	- An invalid Content Owner, Edition, Product, or HSIM value was provided. - The requested Edition and Product are not licensed by your organization. - The requested Annotation ID was not valid. - The requested Annotation ID is not included in the products licensed by your organization.
Authentication or license failed.	Generic error that is returned if the specified hashkey and license information cannot be validated.	Verify the license information and hashkey and resubmit the request.
Citation ID must be a number.	The requested Citation ID contained characters other than numbers	The value provided for the Citation ID contains a value other than the digits 0-9.
Citation was not found.	The requested Citation ID was not found.	- An invalid Content Owner, Edition, or Citation ID value was provided. - The requested Edition is not licensed by your organization. - The requested Citation ID is not included in the products licensed by your organization.
Clinical Indications was not found.	The requested guideline was not found.	- The requested guideline is not of the type expected by the web service. For example, providing an authorization guideline HSIM in an indications for admission request results in this error. - The requested Content Owner, Edition, Product or HSIM code is invalid. - The requested Edition and Product are not licensed by your organization.

Message text	Event	Possible causes
		- Your organization has not licensed the requested edition of the guidelines. - The HSIM code refers to a guideline contained in an unlicensed product.
Response code: 403 Content Not Licensed	The requested guideline was not found.	- The requested Product is not licensed by your organization. - The requested HSIM, Annotation ID, or Citation ID is not valid. - The requested HSIM, Annotation ID, or Citation ID is not contained in any of the products licensed by your organization.
Content owner ' <i>ContentOwnerID</i> ' is unsupported.	The Content Owner value in the request is invalid.	The Content Owner value is not the correct one for your organization.
Definition was not found.	The requested Definition ID was not found.	- An invalid Content Owner, Edition, or Definition ID was provided. - The Definition ID is not valid. - The Definition ID is not contained in the products licensed by your organization.
Edition <i>EditionNumber</i> is not supported for Content Owner <i>ContentOwnerID</i> .	The requested edition of the guidelines is not valid for the specified Content Owner.	- The value for the Content Owner is not valid for your organization. - The Edition number is not valid, or it is not formatted correctly. - The requested edition is not licensed by your organization.
Error deserializing the state.	Generic error that is returned for any problem that occurred while parsing the provided XML or JSON.	Verify the submitted XML or JSON for syntactical errors. Verify that the parameter data, such as an HSIM code, is valid.
<i>Field</i> is a required field.	Required information was not provided in the request.	- Data was not provided for a field in the request. - The provided data is not formatted correctly.
<i>FieldValue</i> is not a valid input type.	An invalid search input was specified.	- When conducting a guideline search based on keyword or code, the Search Type value provided was not valid.
Footnote was not found.	The requested footnote was not found.	- The requested Edition or Product is not licensed by your organization. - The footnote ID is not valid. - The requested footnote is not contained in the products licensed by your organization.

Message text	Event	Possible causes
		- The requested HSIM code is not contained in a product licensed by your organization.
Indications is a required field.	The Indication identifier(s) provided in an Indication POST request were invalid.	- No indication identifiers were specified. - The indication identifiers were not formatted correctly.
No guideline found for guideline code: <i>GuidelineCode</i> .	The requested guideline was not found.	- The Content Owner, Edition, Product, or Guideline Code value was not valid. - The requested Edition or Product is not licensed by your organization.
Product <i>ProductName</i> is not supported.	The requested product is not available.	- The Content Owner or Product name value was not valid. - The requested Product is not licensed by your organization.
<i>Selected</i> is a required field.	The selected flag was not provided or was not recognized.	- Verify that the Selected attribute is present in the answersList data. - Verify that the Selected attribute name is spelled correctly in the answersList data.
State supposed to be in the post body.	An empty POST request was made.	Resubmit the POST request with the required parameter data.
The request is invalid.	The required fields were not provided in the request.	Resubmit the POST request with all required fields.
Recovery Course Stage was not found.	The requested stage of a guideline recovery course was not found.	- The value provided for the Content Owner, Edition, Product, or HSIM field was not valid. - The Recovery Stage value was not numeric. - The requested Recovery Stage does not exist for the specified guideline. - The requested Edition or Product is not licensed by your organization.
Retrieving guideline fails.	The requested guideline was not retrieved.	The most likely cause of this error is that an invalid HSIM code was specified.
Unsupported Care Day.	An invalid recovery stage value was provided.	- The Recovery Stage value was not numeric. - The requested Recovery Stage does not exist for the specified guideline.
Unexpected character encountered while parsing value: <. Path "", line 0, position 0.	The answersList data was provided in an unexpected format.	- Verify that the Parameter content type selected matches the format (JSON or XML) of the answersList data.

Message text	Event	Possible causes								
Uid is a required field.	The UID field was not provided or was not recognized.	- Verify that the Uid attribute is present in the answersList data. - Verify that the Uid attribute name is spelled correctly in the answersList data.								
Unsupported Hsim.	The requested guideline content could not be retrieved.	- The Content Owner, Edition, Product, or HSIM code value was not valid. - The requested Edition or Product is not licensed by your organization. - The requested HSIM code does not exist. - The requested HSIM code is not contained in a product licensed by your organization. - The requested HSIM code is not available using the format of this endpoint.								
Version not supported.	An invalid version of the Cite Care Integrated web service was specified.	Valid values are V1 or V2.								
Response code: 404 (Response body includes html/css to display standard error message).	Unsupported format for the version/edition.	The version/edition number was provided in an unsupported format. The valid formats are: <table><tr><th>Format</th><th>Example</th></tr><tr><td><i>nn</i></td><td>28</td></tr><tr><td><i>nn.nn</i></td><td>27.1 or 27.10</td></tr><tr><td>latest</td><td>latest</td></tr></table> If needed, you can use the following regex to pre-validate the version/edition in a request: <pre>latest (?:\d+(?:\.\d+)?)</pre>	Format	Example	<i>nn</i>	28	<i>nn.nn</i>	27.1 or 27.10	latest	latest
Format	Example									
<i>nn</i>	28									
<i>nn.nn</i>	27.1 or 27.10									
latest	latest									

6 Glossary

The following list describes data elements that commonly appear in the endpoint responses.

Element	Description
ApplicationVersion	Informed by MCG software version number.
BLOS	Benchmark Length of Stay. BLOS types are: A=Adult, G=Geriatric, P=Pediatric. This data appears in search results when available. <pre> BLOS: [{ BlosTypeDisplayValue: "BLOS (0-17)", BlosValue: "A (PO)", BlosType: "BLOS_A" }] </pre>
BlosComments	Optional text field providing more information about a benchmark length of stay (BLOS).
CalculatorTypes	Specifies that a calculator tool is included in the guideline text, and which calculator to use.
CodeDescription	For medical codes, the text description of the code.
CodeType	Type of medical code, such as ICD10-D or DSM-5.
ContentOwner	Type of content edition. This value is usually "GA".
ContentVersion	Edition and version of the care guidelines used for the response.
CrossProduct	Used with the Table of Contents endpoint, this property identifies the product code of the document that the item links to.
CustomGuideline	Indicates whether the guideline used was custom (True = created by the client, False = MCG-provided guideline).
CustomGuidelineDisclaimer	Required disclaimer text displayed with custom guideline content.
DisplayValue	Indicates an identifying value intended for display to a user. For example, in Chronic Care and Transitions of Care assessments, the display value associated with a question might be "Q1".
Edition	Edition of the care guidelines, such as 27th or 28th.
FileId	Used with the Table of Contents endpoint, this property displays the identifier (file name or unique ID) that changes based on the context of the item provided by the LinkType property: Content, Navigation, External, PDF, and Static.
GuidelineCode	Guideline identifier, such as M-5190. The guideline code can be used to locate guidelines in a search but cannot be used to retrieve guideline content.
GuidelineLinks	Displays instances of the links to access other guidelines. Ex: ISC/recoverycourse/0010405100a. <pre> "GuidelineLinks": [{ "ContentVersion": "27.1", "Title": "Cardiac Valve Replacement or Repair", "Hsim": "0010405100a", "Product": "ISC" }] </pre>

Element	Description
GuidelineTitle	Text name of a guideline.
GuidelineType	Identifies the type of content contained in the guideline. Ex: ORG = Optimum Recovery Guideline, or ACG = Ambulatory Care Guideline. See Appendix A for a complete list of guideline types.
Heading	Text name of a guideline section.
HSIM (code)	Health Systems Inventory Management code. A unique guideline identifier. The HSIM code is the primary means used to retrieve guidelines and other content.
Id	Identifies a unique element in the guideline content. The value is an integer stored as a string.
Identified	Used with Chronic Care and Transitions of Care guidelines, this attribute identifies whether a guideline problem is applicable to a patient's plan of care (attribute set to True). Only problems with this attribute set to True are included in the response. Problems with this attribute set to False are omitted from the response.
InputType	Type of data that will be accepted for an individual answer in an assessment. Used with Chronic Care and Transitions of Care guidelines. Values are Date, Numeric, Text, and Mutually Exclusive.
IsDeactivated	Used with the Table of Contents endpoint, this property identifies whether an item is deactivated in the Guideline Modification Module or the Table of Contents Editor.
IsCustom	Used with the Table of Contents endpoint, this property identifies whether an item displays custom content for guidelines/medical policies created using Guideline Modification Module or the Table of Contents Editor.
IsRecoveryMilestone	In an Optimum Recovery Course or General Treatment Course, indicates that an element is a recovery milestone. Recovery milestones are used to determine when the patient is ready to advance to the next day or stage of care.
Items	Used with the Table of Contents endpoint, this property identifies a collection of Item instances representing the structure of the TOC document on the page.
LatestEdition	Most current edition of the care guidelines that you have licensed.
IcdRegions	Optional search parameter for locating Local Coverage Determination (LCD) guidelines in one or more geographic regions. LCD guidelines appear in the Medicare Compliance (MCR) content solution.
LengthOfStay	Optimum length of stay for an inpatient case. Ex: A-2 (DS) = Ambulatory to two days.
LengthOfStayType	Type of length of stay, such as GLOS (Goal Length of Stay) or BLOS (Benchmark Length of Stay).
Link	URL where the text of an element is located.
LinkType	Used with the Table of Contents endpoint, this property determines the context of the item and changes the way the properties are used for rendering items within the TOC structure.
ListLogic	Logic rule associated with a parent indication. The logic rule identifies the number of child indications that must be true for the parent indication to be true. Ex: 1 or more of the following.
MAO	Used with the GET and POST Indication and AuthorizationIndication endpoints, this parameter determines whether to return MAO content in the response. See Optional

Element	Description
	Parameters in Optional parameters for details.
MutuallyExclusive	Property of an answer within an assessment. If set to True, selecting this answer automatically negates any other selected answers for a question. This option applies only to the MultiSelect question type.
Name	In the Bundle endpoint response, the name of an MCG content product.
NoteRequired	Used with the GET and POST Indication endpoints for supplemental Medicare criteria-related indications. This attribute indicates that a note is required and data for it must be provided. The indication cannot be saved without a clinical justification. See the Indication endpoints for details.
Order	Order in which an element will appear.
ParentDisplayName	Displays instances of the parent section name within a content section (such as Care Planning). Ex: ParentDisplayName: Hospitalization
PathId	Identifies instances of an indication's path within a care guideline. The path identifier combined with the indication identifier can be used to identify one or more instances of an indication within a care guideline.
Product	Set of guidelines provided by MCG, such as Chronic Care or RFCRecoveryCourse. To identify your available products, use the Bundle endpoint.
QuestionText	Text of a question in a Chronic Care or Transitions of Care guideline assessment.
ReferenceLink	Indicates that additional details for the guideline can be accessed through another endpoint. This element may appear in various contexts.
Required	Indicates that a field is required and data for it must be provided.
Richtext	This parameter displays all occurrences of formatted text (including bold text, italicized text, font type, and color) within the guideline information as HTML in the response.
SearchDate	Optional filter for filtering guidelines between the effective and termination dates.
SectionSeparator	Used with the Table of Contents endpoint, this property determines whether the current item is immediately preceded by an empty line.
Selected	Used with the Chronic Care and Transitions of Care guidelines, this attribute indicates whether an answer was selected during an assessment. If set to True, the answer was selected. If set to False or not set, the answer has not been selected.
SortOrder	For elements that appear in a series, such as a list of potential answers to a question in an assessment, the order in which the element will appear.
Stage	In a General Treatment Course, the stage of care.
Status	When used with the POST RecoveryStage endpoint, it indicates the patient status based on the submitted care elements and milestones. Values: NotMet (patient is not ready to progress to the next stage); Met (patient is ready to move to the next stage); and Discharge (patient is ready for discharge).
StructuredContentType	Type of content contained in a guideline section, such as "Recovery Course."
Text	Description of an element, such as the text of an annotation, indication, or barrier. The content of this attribute is intended to appear to users.

Element	Description
Title	Text title of a guideline, section, or other guideline component.
TopLevel	For Chronic Care and Transitions of Care guidelines, indicates that the question functions as a header question for a group of questions.
Type	This attribute is used in multiple types of guidelines. For Chronic Care and Transitions of Care, it specifies the type of answer that is required for each question in an assessment. It may also be referred to as the "question type." Available options are SingleSelect, MultiSelect, DEDate, DENumeric, and FreeText. In a plan of care, it specifies the type of intervention to be taken, such as "educate" or "send." In an annotation, Type refers to the type of evidence, such as EVI (Evidence), CRI (Criteria), ALT (Alternative), LOS (Length of Stay), and HOS (Hospitalization).
UID	Identifies a specific set of guideline content within the context of an assessment. For Chronic Care and Transitions of Care guidelines, the UID expands on the ID attribute and includes information about the assessment and section in which the element appears. The last node of the UID is the same value as the ID attribute. The UID attribute is the primary identifier used to retrieve guideline content within the Informed by MCG for Utilization Management web service.
Voice	Used in Chronic Care and Transitions of Care assessments. Indicates the individual, such as the patient or a caregiver, who is expected to answer a question.